2. Interval Analysis

2.1 Introduction

Before using interval analysis as a basic tool in the following chapters, we shall now introduce its main concepts. Section 2.2 recalls fundamental notions on set operators, set functions and set calculus. Section 2.3 then presents basic notions of interval analysis. Section 2.4 is dedicated to the important notion of inclusion function. Finally, Section 2.5 deals with the extension to intervals of logical tests that are almost invariably present in the algorithms of interest to us.

2.2 Operations on Sets

Interval computation is a special case of computation on sets, and set theory provides the foundations for interval analysis. The reader interested only in interval computation may skip this section and go directly to Section 2.3.

The operations on sets fall into two categories. The first one, considered in Section 2.2.1, consists of operations that have a meaning only in a set-theoretic context (such as union, intersection, Cartesian product...). The second one, considered in Section 2.2.2, consists of the extension to sets of operations that are already defined for numbers (or vectors).

2.2.1 Purely set-theoretic operations

Consider two sets $\mathbb{X}$ and $\mathbb{Y}$. Their *intersection* is

$$\mathbb{X} \cap \mathbb{Y} \triangleq \{x \mid x \in \mathbb{X} \text{ and } x \in \mathbb{Y}\}, \tag{2.1}$$

and their *union* is

$$\mathbb{X} \cup \mathbb{Y} \triangleq \{x \mid x \in \mathbb{X} \text{ or } x \in \mathbb{Y}\}. \tag{2.2}$$

$\mathbb{X}$ deprived of $\mathbb{Y}$ is defined by

$$\mathbb{X} \setminus \mathbb{Y} \triangleq \{x \mid x \in \mathbb{X} \text{ and } x \notin \mathbb{Y}\}. \tag{2.3}$$

The *Cartesian product* of $\mathbb{X}$ and $\mathbb{Y}$ is

L. Jaulin et al., *Applied Interval Analysis*
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$$\mathbb{X} \times \mathbb{Y} \triangleq \{(x, y) \mid x \in \mathbb{X} \text{ and } y \in \mathbb{Y}\}. \tag{2.4}$$

If $\mathbb{Z} = \mathbb{X} \times \mathbb{Y}$, then the *projection* of a subset $\mathbb{Z}_1$ of $\mathbb{Z}$ onto $\mathbb{X}$ (with respect to $\mathbb{Y}$) is defined as

$$\text{proj}_{\mathbb{X}}(\mathbb{Z}_1) \triangleq \{x \in \mathbb{X} \mid \exists y \in \mathbb{Y} \text{ such that } (x, y) \in \mathbb{Z}_1\}.$$

These operations are illustrated by Figure 2.1.

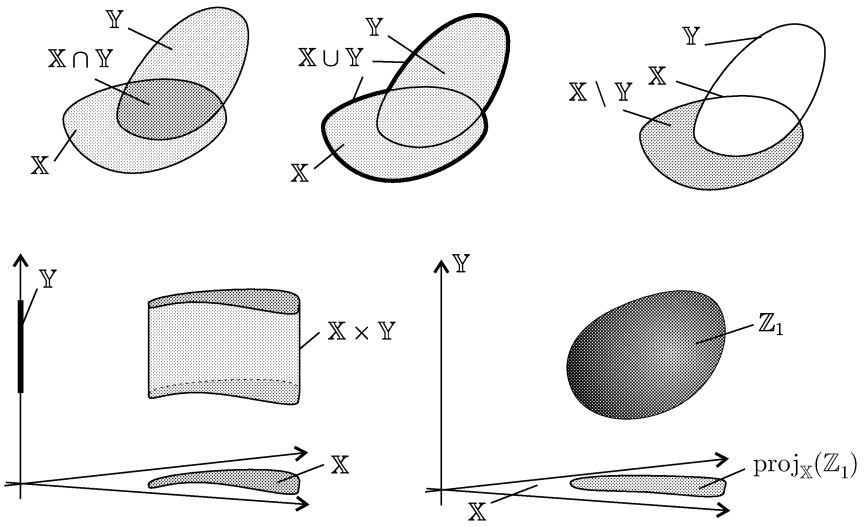


Fig. 2.1. Operations on sets

The *inclusion* of $\mathbb{X}$ in $\mathbb{Y}$ is defined by

$$\mathbb{X} \subset \mathbb{Y} \Leftrightarrow \forall x \in \mathbb{X}, x \in \mathbb{Y}, \tag{2.5}$$

and the *equality* of $\mathbb{X}$ and $\mathbb{Y}$ by

$$\mathbb{X} = \mathbb{Y} \Leftrightarrow \mathbb{X} \subset \mathbb{Y} \text{ and } \mathbb{Y} \subset \mathbb{X}. \tag{2.6}$$

2.2.2 Extended operations

Consider two sets $\mathbb{X}$ and $\mathbb{Y}$ and a function $f : \mathbb{X} \to \mathbb{Y}$. If $\mathbb{X}_1 \subset \mathbb{X}$, the *direct image* of $\mathbb{X}_1$ by f is

$$f(\mathbb{X}_1) \triangleq \{f(x) \mid x \in \mathbb{X}_1\}. \tag{2.7}$$

If $\mathbb{Y}_1 \subset \mathbb{Y}$, the *reciprocal image* of $\mathbb{Y}_1$ by f is

$$f^{-1}(\mathbb{Y}_1) \triangleq \{x \in \mathbb{X} \mid f(x) \in \mathbb{Y}_1\}. \tag{2.8}$$

If $\varnothing$ denotes the empty set, then the previous definitions imply that

$$f(\varnothing) = f^{-1}(\varnothing) = \varnothing. \tag{2.9}$$

It is trivial to show that if $\mathbb{X}_1$ and $\mathbb{X}_2$ are subsets of $\mathbb{X}$ and if $\mathbb{Y}_1$ and $\mathbb{Y}_2$ are subsets of $\mathbb{Y}$, then

$$\begin{aligned}
&f(\mathbb{X}_1 \cap \mathbb{X}_2) \subset f(\mathbb{X}_1) \cap f(\mathbb{X}_2),\\
&f(\mathbb{X}_1 \cup \mathbb{X}_2) = f(\mathbb{X}_1) \cup f(\mathbb{X}_2),\\
&f^{-1}(\mathbb{Y}_1 \cap \mathbb{Y}_2) = f^{-1}(\mathbb{Y}_1) \cap f^{-1}(\mathbb{Y}_2),\\
&f^{-1}(\mathbb{Y}_1 \cup \mathbb{Y}_2) = f^{-1}(\mathbb{Y}_1) \cup f^{-1}(\mathbb{Y}_2),\\
&f\left(f^{-1}(\mathbb{Y})\right) \subset \mathbb{Y},\\
&f^{-1}(f(\mathbb{X})) \supset \mathbb{X},\\
&\mathbb{X}_1 \subset \mathbb{X}_2 \Rightarrow f(\mathbb{X}_1) \subset f(\mathbb{X}_2),\\
&\mathbb{Y}_1 \subset \mathbb{Y}_2 \Rightarrow f^{-1}(\mathbb{Y}_1) \subset f^{-1}(\mathbb{Y}_2),\\
&\mathbb{X} \subset \mathbb{Y}_1 \times \mathbb{Y}_2 \Rightarrow \mathbb{X} \subset \mathrm{proj}_{\mathbb{Y}_1}(\mathbb{X}) \times \mathrm{proj}_{\mathbb{Y}_2}(\mathbb{X}).
\end{aligned} \tag{2.10}$$

In the same manner, it is possible to extend operations on numbers (or vectors) to operations on sets. Denote by $\mathcal{P}(\mathbb{X})$ the *power set* of $\mathbb{X}$, *i.e.*, the set of all subsets of $\mathbb{X}$. Let $\diamond$ be a binary operator from $\mathbb{X} \times \mathbb{Y}$ to $\mathbb{Z}$. It can be extended as a set operator as follows:

$$\mathbb{X}_1 \diamond \mathbb{Y}_1 \triangleq \{x_1 \diamond y_1 \mid x_1 \in \mathbb{X}_1, y_1 \in \mathbb{Y}_1\}, \tag{2.11}$$

where $\diamond$ is now an operator from $\mathcal{P}(\mathbb{X} \times \mathbb{Y})$ to $\mathcal{P}(\mathbb{Z})$. For instance, if $\mathbb{X}_1$ and $\mathbb{Y}_1$ are subsets of $\mathbb{R}^n$, then

$$\mathbb{X}_1 + \mathbb{Y}_1 = \{\mathbf{x} + \mathbf{y} \mid \mathbf{x} \in \mathbb{X}_1, \mathbf{y} \in \mathbb{Y}_1\}, \tag{2.12}$$

$$\mathbb{X}_1 - \mathbb{Y}_1 = \{\mathbf{x} - \mathbf{y} \mid \mathbf{x} \in \mathbb{X}_1, \mathbf{y} \in \mathbb{Y}_1\}. \tag{2.13}$$

Note that the set $\mathbb{X}_1 - \mathbb{X}_1 = \{\mathbf{x} - \mathbf{y} \mid \mathbf{x} \in \mathbb{X}_1, \mathbf{y} \in \mathbb{X}_1\}$ should not be confused with the set $\{\mathbf{x} - \mathbf{x} \mid \mathbf{x} \in \mathbb{X}_1\} = \{\mathbf{0}\}$.

When the operator $\diamond$ applies to an element x_1 of $\mathbb{X}$ together with a subset $\mathbb{Y}_1$ of $\mathbb{Y}$, x_1 is cast into the singleton $\{x_1\}$, so

$$x_1 \diamond \mathbb{Y}_1 \triangleq \{x_1\} \diamond \mathbb{Y}_1 = \{x_1 \diamond y_1 \mid y_1 \in \mathbb{Y}_1\}. \tag{2.14}$$

For instance, if $\mathbb{D}$ is the disk of $\mathbb{R}^2$ with centre $\mathbf{c}$ and radius 4, then $3 * \mathbb{D}$ is the disk of $\mathbb{R}^2$ with centre $3\mathbf{c}$ and radius 12. If $\mathbb{Y}_1$ is also a singleton $\{y_1\}$, then

$$x_1 \diamond y_1 = \{x_1\} \diamond \{y_1\} = \{x_1 \diamond y_1\}, \tag{2.15}$$

and the usual rules of $\diamond$ for punctual arguments apply.

2.2.3 Properties of set operators

Some properties of operators acting on numbers extend to their set counterparts. Consider, as an illustration, a set $\mathbb{X}$ equipped with the binary operator $\diamond$. Assume that $\mathbb{X}$ is *closed* with respect to $\diamond$ (*i.e.*, if x and y belong to $\mathbb{X}$,

then $x \diamond y$ belongs to $\mathbb{X}$). Assume also that $\diamond$ has been extended to $\mathcal{P}(\mathbb{X})$ as described in the previous section. Some properties true for $(\mathbb{X}, \diamond)$ remain true for $(\mathcal{P}(\mathbb{X}), \diamond)$. For instance, if $\diamond$ is commutative for $\mathbb{X}$, it is also commutative for $\mathcal{P}(\mathbb{X})$, *i.e.*,

$$\begin{aligned} &(\forall x_1 \in \mathbb{X}, \forall x_2 \in \mathbb{X}, x_1 \diamond x_2 = x_2 \diamond x_1) \\ \Rightarrow\ &(\forall \mathbb{X}_1 \in \mathcal{P}(\mathbb{X}), \forall \mathbb{X}_2 \in \mathcal{P}(\mathbb{X}), \mathbb{X}_1 \diamond \mathbb{X}_2 = \mathbb{X}_2 \diamond \mathbb{X}_1)\,. \end{aligned} \tag{2.16}$$

If $\diamond$ is associative for $\mathbb{X}$, it is also associative for $\mathcal{P}(\mathbb{X})$, *i.e.*,

$$\begin{aligned} &\left(\forall\,(x_1, x_2, x_3) \in \mathbb{X}^3, x_1 \diamond (x_2 \diamond x_3) = (x_1 \diamond x_2) \diamond x_3\right) \\ \Rightarrow\ &\left(\forall\,(\mathbb{X}_1, \mathbb{X}_2, \mathbb{X}_3) \in (\mathcal{P}(\mathbb{X}))^3\,, \mathbb{X}_1 \diamond (\mathbb{X}_2 \diamond \mathbb{X}_3) = (\mathbb{X}_1 \diamond \mathbb{X}_2) \diamond \mathbb{X}_3\right). \end{aligned} \tag{2.17}$$

If $\diamond$ admits a unit element (denoted by 0) in $\mathbb{X}$, then it also admits a unit element in $\mathcal{P}(\mathbb{X})$, which is the singleton $\{0\}$.

On the other hand, some properties true for $(\mathbb{X}, \diamond)$ may become false for $(\mathcal{P}(\mathbb{X}), \diamond)$. For instance, if each element x of $(\mathbb{X}, \diamond)$ admits a symmetric element this is no longer true in $(\mathcal{P}(\mathbb{X}), \diamond)$. We only have

$$\begin{aligned} &(\forall x_1 \in \mathbb{X}, \exists y_1 \in \mathbb{X} \mid x_1 \diamond y_1 = 0) \\ \Rightarrow\ &(\forall \mathbb{X}_1 \in \mathcal{P}(\mathbb{X}), \exists \mathbb{Y}_1 \in \mathcal{P}(\mathbb{X}) \mid \mathbb{X}_1 \diamond \mathbb{Y}_1 \ni 0)\,. \end{aligned} \tag{2.18}$$

Thus, if $(\mathbb{X}, \diamond)$ is a group, $(\mathcal{P}(\mathbb{X}), \diamond)$ is only a monoid.

It is not our purpose here to give an exhaustive view of the extensions of operators on numbers to sets, but just to stress that special care should be exercised when dealing with set arithmetic. For instance, if $\mathbb{X}_1$ is a subset of a group $(\mathbb{X}, +)$, an expression such as

$$\mathbb{Z} = \mathbb{X}_1 - \mathbb{X}_1 \tag{2.19}$$

should not be interpreted as

$$\mathbb{Z} = \{\mathbf{x} - \mathbf{x} \mid \mathbf{x} \in \mathbb{X}_1\}\,, \tag{2.20}$$

but as

$$\mathbb{Z} = \{\mathbf{x} - \mathbf{y} \mid \mathbf{x} \in \mathbb{X}_1, \mathbf{y} \in \mathbb{X}_1\}, \tag{2.21}$$

see (2.13). To avoid such confusions, the use of algebraic manipulations on sets will be limited as much as possible in this book.

The following theorem will have important consequences in the context of interval computation.

Theorem 2.1 *Consider the function*

$$f : \begin{array}{l} \mathbb{X}(1) \times \cdots \times \mathbb{X}(n) \to \mathbb{Y} \\ (x_1, \ldots, x_n) \qquad\quad \mapsto y \end{array}$$

for which a formal expression involving operators and functions is available. Let $\mathbb{F}(\mathbb{X}_1, \ldots, \mathbb{X}_n)$ *be a function from* $\mathcal{P}(\mathbb{X}(1)) \times \cdots \times \mathcal{P}(\mathbb{X}(n))$ *to* $\mathcal{P}(\mathbb{Y})$*. Assume that* $\mathbb{F}(\mathbb{X}_1, \ldots, \mathbb{X}_n)$ *has the same formal expression as* $f(\mathbb{X}_1, \ldots, \mathbb{X}_n)$*. Then*

$$f(\mathbb{X}_1, \ldots, \mathbb{X}_n) \subset \mathbb{F}(\mathbb{X}_1, \ldots, \mathbb{X}_n), \tag{2.22}$$

where

$$f(\mathbb{X}_1, \ldots, \mathbb{X}_n) = \{f(x_1, \ldots, x_n) \mid x_1 \in \mathbb{X}_1, \ldots, x_n \in \mathbb{X}_n\}.$$

Moreover, if each x_i *occurs at most once in the formal expression of* f*, then*

$$f(\mathbb{X}_1, \ldots, \mathbb{X}_n) = \mathbb{F}(\mathbb{X}_1, \ldots, \mathbb{X}_n). \tag{2.23}$$

■

A proof of Theorem 2.1 in the case where all functions involved in the formal expression of f are continuous and where the sets $\mathbb{X}_i$ are closed intervals may be found in Moore (1979). The extension to more general sets (as presented in Theorem 2.1) is a direct consequence of the propagation theorem (Jaulin, Kieffer, Braems and Walter, 2001), the principle of which will be recalled in Section 6.4.4, page 174.

In general, unfortunately, $\mathbb{F}(\mathbb{X}_1, \ldots, \mathbb{X}_n)$ is only an outer approximation of $f(\mathbb{X}_1, \ldots, \mathbb{X}_n)$, because of multiple occurrences of variables in the formal expression of f. Set computation is thus *pessimistic*, because of the *dependency effect* illustrated by the following example.

Example 2.1 *Consider the function* $f : \mathbb{R}^2 \to \mathbb{R}$*,* $f(x_1, x_2) = x_1 + x_2 - x_1$*. Then*

$$\begin{aligned} f(\mathbb{X}_1, \mathbb{X}_2) &= \{x_1 + x_2 - x_1 \mid x_1 \in \mathbb{X}_1, x_2 \in \mathbb{X}_2\} \\ &= \{x_2 \mid x_2 \in \mathbb{X}_2\} = \mathbb{X}_2, \end{aligned} \tag{2.24}$$

and

$$\begin{aligned} \mathbb{F}(\mathbb{X}_1, \mathbb{X}_2) &= \mathbb{X}_1 + \mathbb{X}_2 - \mathbb{X}_1 \\ &= \{x_1 + x_2 - x_3 \mid x_1 \in \mathbb{X}_1, x_2 \in \mathbb{X}_2, x_3 \in \mathbb{X}_1\}. \end{aligned} \tag{2.25}$$

It is clear that $f(\mathbb{X}_1, \mathbb{X}_2) \subset \mathbb{F}(\mathbb{X}_1, \mathbb{X}_2)$*. The dependency between* x_1 *and* x_3 *(*$x_1 = x_3$*) has been forgotten in (2.25), thus adding one degree of freedom in the elaboration of the set* $\mathbb{F}(\mathbb{X}_1, \mathbb{X}_2)$*; hence the pessimism.* ■

Note that multiple occurrences of variables associated to singletons do not cause pessimism.

2.2.4 Wrappers

Consider a set $\mathbb{X}$ and a set $\mathbb{IX}$ of subsets of $\mathbb{X}$. $\mathbb{IX}$ is a *set of wrappers* for $\mathbb{X}$ if $\mathbb{X}$ and each singleton of $\mathbb{X}$ belong to $\mathbb{IX}$ and if $\mathbb{IX}$ is closed by intersection (*i.e.*, if $\mathbb{X}_1 \in \mathbb{IX}$ and $\mathbb{X}_2 \in \mathbb{IX}$, then $\mathbb{X}_1 \cap \mathbb{X}_2 \in \mathbb{IX}$). The empty set $\varnothing$ must thus belong to $\mathbb{IX}$, unless $\mathbb{X}$ is a singleton.

Example 2.2 *A set of wrappers for* $\mathbb{X} = \{a, b, c, d\}$ *is*

$$\mathbb{IX} = \{\varnothing, a, b, c, \{a, b\}, \{a, d\}, \{a, b, c, d\}\}. \tag{2.26}$$

■

Let $\mathbb{X}_1$ be a subset of $\mathbb{X}$; the *smallest wrapper* $[\mathbb{X}_1]$ of $\mathbb{X}_1$ is the smallest element of $\mathbb{IX}$ containing $\mathbb{X}_1$:

$$[\mathbb{X}_1] = \bigcap \{\mathbb{Y} \in \mathbb{IX} \mid \mathbb{X}_1 \subset \mathbb{Y}\}. \tag{2.27}$$

Example 2.3 *The set* $\mathcal{P}(\mathbb{X})$ *of all subsets of* $\mathbb{X}$ *is a set of wrappers for* $\mathbb{X}$. *If* $\mathbb{X}_1$ *is a subset of* $\mathbb{X}$, *then* $[\mathbb{X}_1] = \mathbb{X}_1$. ■

Example 2.4 *When* $\mathbb{X} = \mathbb{R}^n$, *the set* $\mathbb{IX}$ *of all convex sets of* $\mathbb{R}^n$ *is a set of wrappers for* $\mathbb{X}$. $[\mathbb{X}_1]$ *is then the convex hull of* $\mathbb{X}_1$. ■

In practice, *wrappers* should be simple enough to allow the computation of outer approximations of sets. The wrappers to be considered in this book are intervals of $\mathbb{R}$, axis-aligned boxes of $\mathbb{R}^n$, subsets of the Boolean set {true, false} and unions of intervals or boxes. Other types of wrapper could be used, such as parallelotopes, zonotopes, convex sets or unions of such sets. Although they are not closed with respect to intersection, ellipsoids can also be used (Milanese et al., 1996).

Let $\diamond$ be a binary operator from $\mathbb{X} \times \mathbb{Y}$ to $\mathbb{Z}$. Let $\mathbb{IX}$, $\mathbb{IY}$ and $\mathbb{IZ}$ be sets of wrappers for $\mathbb{X}$, $\mathbb{Y}$ and $\mathbb{Z}$. The operator $\diamond$ can be extended to these sets of wrappers as follows. If $\mathbb{X}_1 \in \mathbb{IX}$ and $\mathbb{Y}_1 \in \mathbb{IY}$, then

$$\mathbb{X}_1 \ [\diamond] \ \mathbb{Y}_1 \triangleq [\{x_1 \diamond y_1 \mid x_1 \in \mathbb{X}_1, y_1 \in \mathbb{Y}_1\}], \tag{2.28}$$

where the wrapping operator [.] on the right-hand side is defined as in (2.27). Because of the definition of $\mathbb{X}_1 \diamond \mathbb{Y}_1$ given in (2.11),

$$\mathbb{X}_1 \ [\diamond] \ \mathbb{Y}_1 \ \supset \ \mathbb{X}_1 \diamond \mathbb{Y}_1. \tag{2.29}$$

Consider two sets $\mathbb{X}$ and $\mathbb{Y}$, a function $f : \mathbb{X} \to \mathbb{Y}$, and two sets of wrappers $\mathbb{IX}$ and $\mathbb{IY}$ for $\mathbb{X}$ and $\mathbb{Y}$ respectively. If $\mathbb{X}_1 \in \mathbb{IX}$, then f can be extended to wrappers as

$$[f]([\mathbb{X}_1]) \triangleq [\{f(x) \mid x \in \mathbb{X}_1\}]. \tag{2.30}$$

Again,

$$[f]([\mathbb{X}_1]) \supset f([\mathbb{X}_1]). \tag{2.31}$$

Consider three sets $\mathbb{X}$, $\mathbb{Y}$, $\mathbb{Z}$, their sets of wrappers $\mathbb{IX}$, $\mathbb{IY}$, $\mathbb{IZ}$ and three functions $f : \mathbb{X} \to \mathbb{Y}$, $g : \mathbb{Y} \to \mathbb{Z}$, $h = g \circ f$ (where $\circ$ denotes the composition operator). Then, $g(f([\mathbb{X}_1])) = h([\mathbb{X}_1])$, but one can only write that

$$[g]([f]([\mathbb{X}_1])) \supset [h]([\mathbb{X}_1]). \tag{2.32}$$

This is known as the *wrapping effect*, illustrated by Figure 2.2 when the wrappers are convex sets of $\mathbb{R}^2$.

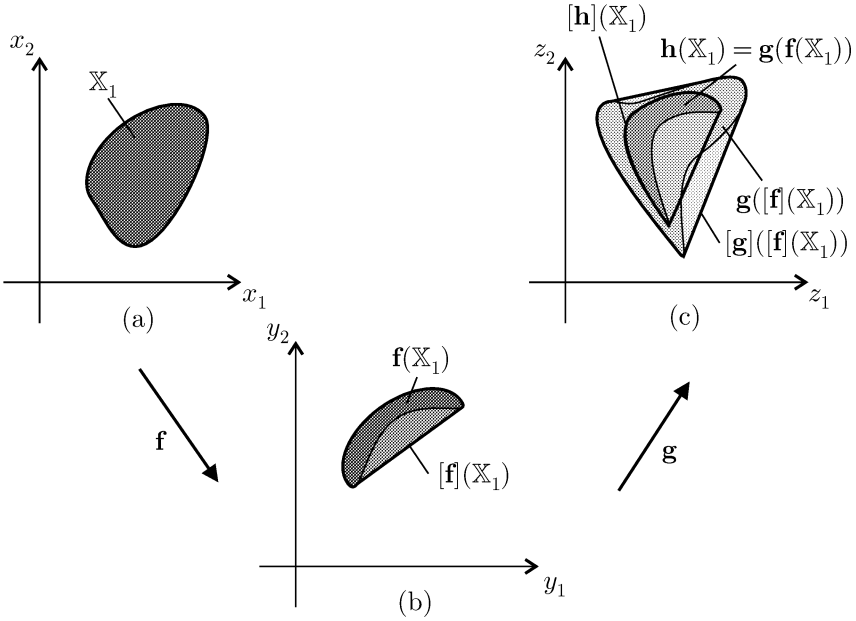


Fig. 2.2. Wrapping effect when the wrappers are the convex sets of $\mathbb{R}^2$; (a) $\mathbb{X}_1$ is convex and thus a wrapper; (b) $\mathbf{f}(\mathbb{X}_1)$, in dark grey, is not convex; its convex hull is the wrapper $[\mathbf{f}](\mathbb{X}_1)$; (c) since $\mathbf{f}(\mathbb{X}_1) \subset [\mathbf{f}](\mathbb{X}_1)$, we have $\mathbf{g}(\mathbf{f}(\mathbb{X}_1)) \subset \mathbf{g}([\mathbf{f}](\mathbb{X}_1)) \subset [\mathbf{g}]([\mathbf{f}](\mathbb{X}_1))$; therefore, $[\mathbf{g}]([\mathbf{f}](\mathbb{X}_1))$ is a convex set that contains the convex hull $[\mathbf{h}](\mathbb{X}_1)$ of $\mathbf{g}(\mathbf{f}(\mathbb{X}_1))$; the undesirable points of $[\mathbf{g}]([\mathbf{f}](\mathbb{X}_1))$ that are outside $[\mathbf{h}](\mathbb{X}_1)$ are a direct consequence of the wrapping effect

As the dependency effect, the wrapping effect introduces pessimism when computing with wrappers. Some properties that held true for set computation are thus no longer true for computation with wrappers. This is the case for Theorem 2.1; when wrappers are used, the inclusion property (2.22) is still satisfied, but does not necessarily transform into the equality (2.23) when each input variable occurs at most once in the formal expression of f.

Remark 2.1 *In this book, computation will be with wrappers and not with generic sets. When this creates no ambiguity,* $\mathbb{X}_1 \,[\diamond]\, \mathbb{Y}_1$ *will be shortened into* $\mathbb{X}_1 \diamond \mathbb{Y}_1$. ■

2.3 Interval Analysis

In interval analysis, the wrappers to be used are intervals when dealing with $\mathbb{R}$ and axis-aligned boxes when dealing with $\mathbb{R}^n$. Using the concepts of set computation recalled in Section 2.2, we shall now present interval computation.

2.3.1 Intervals

An *interval real* $[x]$ is a connected subset of $\mathbb{R}$. Even when the interval is not closed, we shall keep to the notation $[x]$. When no confusion may arise, $[x]$ will often merely be called an *interval*. Whether the empty set $\varnothing$ should be considered as an interval is still a subject of discussion. We choose to answer positively, if only to ensure that the set of intervals is closed with respect to intersection, and also because $\varnothing$ represents the absence of solution of a problem. The *lower bound* $\mathrm{lb}([x])$ of an interval $[x]$, also denoted by $\underline{x}$, is defined as

$$\underline{x} = \mathrm{lb}([x]) \triangleq \sup\{a \in \mathbb{R} \cup \{-\infty, \infty\} \mid \forall x \in [x], a \leqslant x\}. \tag{2.33}$$

Its *upper bound* $\mathrm{ub}([x])$, also denoted by $\overline{x}$, is defined as

$$\overline{x} = \mathrm{ub}([x] \triangleq \inf\{b \in \mathbb{R} \cup \{-\infty, \infty\} \mid \forall x \in [x], x \leqslant b\}. \tag{2.34}$$

Thus, $\underline{x}$ is the largest number on the left of $[x]$ and $\overline{x}$ is the smallest number on its right. For instance if $[x] =]-3, 7]$ then $\underline{x} = -3$ and $\overline{x} = 7$; if $[x] =]-\infty, \infty[$ then $\underline{x} = -\infty$ and $\overline{x} = \infty$. The *width* of any non-empty interval $[x]$ is

$$w([x]) \triangleq \overline{x} - \underline{x}, \tag{2.35}$$

so $w(]3, \infty[) = \infty$. The *midpoint* (or *centre*) of any bounded and non-empty interval $[x]$ is defined as

$$\mathrm{mid}([x]) \triangleq \frac{\underline{x} + \overline{x}}{2}. \tag{2.36}$$

The set-theoretic operations of Section 2.2.1 can be applied to intervals. The *intersection* of two intervals $[x]$ and $[y]$, defined by

$$[x] \cap [y] \triangleq \{z \in \mathbb{R} \mid z \in [x] \text{ and } z \in [y]\}, \tag{2.37}$$

is always an interval. This is not the case for their *union*

$$[x] \cup [y] \triangleq \{z \in \mathbb{R} \mid z \in [x] \text{ or } z \in [y]\}. \tag{2.38}$$

To make the set of intervals closed with respect to union, define the *interval hull* of a subset $\mathbb{X}$ of $\mathbb{R}$ as the smallest interval $[\mathbb{X}]$ that contains it. This is consistent with (2.27). For instance, the interval hull of $]2, 3] \cup [5, 7]$ is the interval $]2, 7]$. Define the *interval union* of $[x]$ and $[y]$, denoted by $[x] \sqcup [y]$, as the interval hull of $[x] \cup [y]$, *i.e.*,

$$[x] \sqcup [y] \triangleq [[x] \cup [y]]. \tag{2.39}$$

In the same manner,

$$[x] \ [\backslash] \ [y] = [[x] \setminus [y]] = [\{x \in [x] \mid x \notin [y]\}]. \tag{2.40}$$

For instance, $[0, 5[\ [\backslash] \ ]3, 4[= [0, 5[$ and $[0, 5[\ [\backslash] \ ]3, 5[= [0, 3]$. The Cartesian product of two intervals is not an interval but a box of $\mathbb{R}^2$; it thus corresponds to an external operation, to be treated in Section 2.3.4.

2.3.2 Interval computation

The four classical operations of real arithmetic, namely addition ($+$), subtraction ($-$), multiplication ($*$) and division ($/$) can be extended to intervals. For any such binary operator, denoted by $\diamond$, performing the operation associated with $\diamond$ on the intervals $[x]$ and $[y]$ means computing

$$[x] \diamond [y] = [\{x \diamond y \in \mathbb{R} \mid x \in [x], y \in [y]\}], \tag{2.41}$$

which is a direct consequence of (2.28). For instance

$$(]1, 2.2] * [0, 2[) +]1, 3] = [0, 4.4[+]1, 3] =]1, 7.4[, \tag{2.42}$$

$$1/[-2, 2[=] - \infty, \infty[, \tag{2.43}$$

$$[3, 4]/[0, 0] = \varnothing. \tag{2.44}$$

The rule (2.41) was first presented in the context of bounded and closed intervals by Moore (1959) and then extended to open-ended unbounded intervals (Hanson, 1968; Kahan, 1968; Davis, 1987).

Remark 2.2 *The result of (2.44) is based on a mathematical interpretation of the interval* $[0, 0]$*. We shall see in Chapter 10 that two types of zero are distinguished in the floating-point representation to be used when implementing interval computation, which sometimes allow different results.* ■

Remark 2.3 *Definition (2.41) differs from that given for sets in (2.11) because it is required that* $\diamond$ *returns an interval and not a possibly non-connected subset of* $\mathbb{R}$*. Because of wrapping, we should have written* $[x]$ $[\diamond]$ $[y]$ *instead of* $[x] \diamond [y]$ *but, as for generic wrappers, we shall keep to the simpler notation* $[x] \diamond [y]$*. Note that (2.41) and (2.11) match when* $\diamond$ *is continuous, as the set* $\{x \diamond y \in \mathbb{R} \mid x \in [x], y \in [y]\}$ *is then an interval.* ■

Remark 2.4 *The operator* $\diamond$ *defined for intervals can also be extended to unions of intervals (also called* discontinuous intervals*) (Hyvönen, 1992). For instance, one may write*

$$1/[-2, 2[=] - \infty, -1/2[\cup [1/2, \infty[, \tag{2.45}$$

instead of $]-\infty, \infty[$ *as in (2.43). The computation on unions of intervals may, however, lead to an explosion of the number of subintervals to be handled and will no longer be considered in this book.* ■

As already mentioned for generic sets, the properties of the basic operators for intervals differ from their properties in $\mathbb{R}$. For instance, $[x]-[x]$ is generally not equal to $[0, 0]$. This is because $[x] - [x] = \{x - y \mid x \in [x], y \in [x]\}$, rather than $\{x - x \mid x \in [x]\}$. The subtraction thus does not take the dependency of the two occurrences of $[x]$ into account. Addition and multiplication remain associative and commutative, but multiplication is no longer distributive with respect to addition. Instead,

$$[x] * ([y] + [z]) \subset [x] * [y] + [x] * [z], \tag{2.46}$$

a property known as *subdistributivity*, which is a direct consequence of the dependency effect, as $[x]$ appears only once on the left-hand side but twice on the right-hand side. As a result, it is recommended to factorize expanded forms as much as possible.

Elementary functions such as exp, tan, sin, cos... extend to intervals as indicated by (2.30). If f is a function from $\mathbb{R}$ to $\mathbb{R}$, then its interval counterpart $[f]$ satisfies

$$[f]([x]) = [\{f(x) \mid x \in [x]\}]\,. \tag{2.47}$$

For any continuous elementary function, $[f]([x])$ is thus equal to the image set $f([x])$. For instance,

$$\begin{aligned}
&[\arctan]([0,\infty[) = [0,\pi/2[,\\
&[\mathrm{sqr}]([-1,3[) = [0,9[,\\
&[\exp]([0,1[) = [\exp(0),\exp(1)[= [1,e[,\\
&[\mathrm{sqrt}]([4,25]) = [\mathrm{sqrt}(4),\ \mathrm{sqrt}(25)] = [2,5],\\
&[\mathrm{sqrt}]([-25,-4]) = \varnothing,\\
&[\mathrm{sqrt}]([-50,1]) = [\mathrm{sqrt}]([0,1]) = [0,1],\\
&[\ln]([-50,1[) =]-\infty,0[\,.
\end{aligned} \tag{2.48}$$

2.3.3 Closed intervals

This section presents rules for computing on subsets of $\mathbb{R}$ in the special case where the wrappers are closed intervals of $\mathbb{R}$. Denote by $\mathbb{IR}$ the set of all such closed intervals. Since $\mathbb{R}$ and $\varnothing$ are both open and closed, they both belong to $\mathbb{IR}$, and any element of $\mathbb{IR}$ can be written in one of the following forms: $[a,b]$, $]-\infty,a]$, $[a,\infty[$, $]-\infty,\infty[$ or $\varnothing$, where a and b are real numbers such that $a \leqslant b$. Any $[x]$ of $\mathbb{IR}$ can be specified in a unique way by its lower bound $\underline{x}$ and its upper bound $\overline{x}$. For simplicity, we shall often write $[x] = [\underline{x},\overline{x}]$, even if bounds may be infinite. Thus, $[0,\infty]$ should be interpreted as $[0,\infty[$. Note the dual nature of closed intervals, which may be viewed as *sets* (on which standard set operations apply), and as *couples of elements* of $\mathbb{R}$ on which an arithmetic can be built. Couples of the form $[\infty,\infty]$, $[-\infty,-\infty]$ and $[a,b]$ with $a > b$ do not correspond to intervals (see *modal interval arithmetic* (Gardenes et al., 1985), however, where the notation $[a,b]$ with $a > b$ receives a meaning). When $\underline{x}$ and $\overline{x}$ are equal, the interval $[x]$ is said to be *punctual* (or degenerate). Any real number could thus be represented as a punctual interval and vice versa.

The operations of Section 2.3.2 can be redefined, in the context of closed intervals, as operations on their bounds: the bounds of the result of an interval operation are expressed as functions of the bounds of its interval arguments. The remainder of this section is devoted to illustrating how this can be done.

The interval union of two non-empty closed intervals $[x]$ and $[y]$, defined by (2.39), satisfies

$$\forall [x] \in \mathbb{IR}, \forall [y] \in \mathbb{IR}, \quad [x] \sqcup [y] = \left[\min\{\underline{x}, \underline{y}\}, \max\{\overline{x}, \overline{y}\}\right]. \tag{2.49}$$

The intersection of two non-empty closed intervals $[x]$ and $[y]$, defined by (2.37), satisfies

$$\begin{aligned} [x] \cap [y] &= \left[\max\{\underline{x}, \underline{y}\}, \min\{\overline{x}, \overline{y}\}\right] \text{ if } \max\{\underline{x}, \underline{y}\} \leqslant \min\{\overline{x}, \overline{y}\}, \\ &= \varnothing \text{ otherwise.} \end{aligned} \tag{2.50}$$

If α is a real number and $[x]$ a non-empty interval, then the interval

$$\alpha[x] \triangleq \{\alpha x \mid x \in [x]\} \tag{2.51}$$

is given by

$$\begin{aligned} \alpha[x] &= [\alpha\underline{x}, \alpha\overline{x}] \quad \text{if } \alpha \geqslant 0 \\ &= [\alpha\overline{x}, \alpha\underline{x}] \quad \text{if } \alpha < 0. \end{aligned} \tag{2.52}$$

For non-empty closed intervals,

$$\begin{aligned} [x] + [y] &= [\underline{x} + \underline{y}, \overline{x} + \overline{y}], \\ [x] - [y] &= [\underline{x} - \overline{y}, \overline{x} - \underline{y}], \\ [x] * [y] &= [\min\{\underline{x}\underline{y}, \underline{x}\overline{y}, \overline{x}\underline{y}, \overline{x}\overline{y}\}, \max\{\underline{x}\underline{y}, \underline{x}\overline{y}, \overline{x}\underline{y}, \overline{x}\overline{y}\}]. \end{aligned} \tag{2.53}$$

The product of two intervals will be denoted indifferently by $[x]*[y]$, or $[x][y]$.

For division, (2.41) leads to

$$\begin{aligned} 1/[y] &= \varnothing && \text{if } [y] = [0, 0], \\ &= [1/\overline{y}, 1/\underline{y}] && \text{if } 0 \notin [y], \\ &= [1/\overline{y}, \infty[&& \text{if } \underline{y} = 0 \text{ and } \overline{y} > 0, \\ &=]-\infty, 1/\underline{y}] && \text{if } \underline{y} < 0 \text{ and } \overline{y} = 0, \\ &=]-\infty, \infty[&& \text{if } \underline{y} < 0 \text{ and } \overline{y} > 0, \end{aligned} \tag{2.54}$$

and

$$[x]/[y] = [x] * (1/[y]). \tag{2.55}$$

Of course, when applied to punctual intervals $[x]$ and $[y]$, the previous rules simplify into the usual rules of real arithmetic, which is why interval arithmetic can claim to be an extension of the latter.

Elementary interval functions can also be expressed in terms of bounds. For instance, for any non-empty $[x]$,

$$[\exp]([x]) = [\exp(\underline{x}), \exp(\overline{x})]. \tag{2.56}$$

For non-monotonic functions, the situation is more complicated. For example, $[\sin]([0, \pi]) = [0, 1]$ differs from the interval $[\sin(0), \sin(\pi)] = [0, 0]$. Specific

Table 2.1. Algorithm for the interval evaluation of the sine function

Algorithm sin(in: $[x]$; out: $[r]$)	
1	if $\exists k \in \mathbb{Z} \mid 2k\pi - \pi/2 \in [x]$ then $\underline{r} = -1$;
2	else $\underline{r} = \min(\sin \underline{x}, \sin \overline{x})$;
3	if $\exists k \in \mathbb{Z} \mid 2k\pi + \pi/2 \in [x]$ then $\overline{r} = 1$;
4	else $\overline{r} = \max(\sin \underline{x}, \sin \overline{x})$.

algorithms must therefore be built. An algorithm for the sine function is given by Table 2.1.

Figure 2.3 illustrates the computation of $\sin([2.6, 7.2]) = [-1, 0.7937]$. Inclusion functions for other trigonometric functions are obtained in the same way or by expressing them as functions of the sine function. The hyperbolic functions receive a similar treatment.

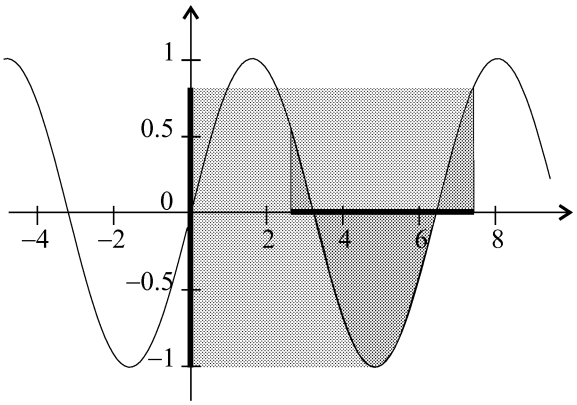


Fig. 2.3. Computation of $\sin([x])$

Remark 2.5 *Most often, an operation or the evaluation of a function with an empty interval argument should yield an empty result. Obvious exceptions to this rule are for the union operator* $\cup$ *and interval union operator* $\sqcup$*, as* $[x] \cup \varnothing = [x]$ *and* $[x] \sqcup \varnothing = [x]$ ■

As already mentioned, the arithmetical rules for intervals differ from those for real numbers. For instance, $x^2 - x = (x - \frac{1}{2})^2 - \frac{1}{4}$ whereas $[x]^2 - [x]$ differs from $([x] - \frac{1}{2})^2 - \frac{1}{4}$, as illustrated by the following example.

Example 2.5 *At* $[x] = [-1, 3]$,

$$[x]^2 - [x] = [-1, 3]^2 - [-1, 3] = [0, 9] + [-3, 1] = [-3, 10], \tag{2.57}$$

$$\left([x] - \tfrac{1}{2}\right)^2 - \tfrac{1}{4} = \left[-\tfrac{3}{2}, \tfrac{5}{2}\right]^2 - \tfrac{1}{4} = \left[0, \tfrac{25}{4}\right] - \tfrac{1}{4} = \left[-\tfrac{1}{4}, 6\right]. \quad (2.58)$$

The first result is a pessimistic approximation of the image set of $x^2 - x$ *at* $[-1, 3]$*, whereas the second one is equal to this image set (see Theorem 2.1).* ■

One is therefore well advised to transform expressions in such a way as to reduce the number of occurrences of each variable as much as possible, which is sometimes easier said than done.

Since computation on closed intervals reduces to computation on their bounds, traditional interval software considers only closed intervals (except packages such as INC++ or PROLOG 4, which also support open and half-open intervals, at the cost of a much more complex implementation).

2.3.4 Interval vectors

An *interval real vector* $[\mathbf{x}]$ is a subset of $\mathbb{R}^n$ that can be defined as the Cartesian product of n closed intervals. When there is no ambiguity, $[\mathbf{x}]$ will simply be called an interval vector, or a *box*. It will be written as

$$[\mathbf{x}] = [x_1] \times [x_2] \times \cdots \times [x_n], \text{ with } [x_i] = [\underline{x}_i, \overline{x}_i] \text{ for } i = 1, \ldots, n. \quad (2.59)$$

Its ith *interval component* $[x_i]$ is the projection of $[\mathbf{x}]$ onto the ith axis. The empty set of $\mathbb{R}^n$ should thus be written as $\varnothing \times \cdots \times \varnothing$ because all of its interval components are empty. Expressions such as

$$[\mathbf{x}] = \varnothing \times [0, 1] \quad (2.60)$$

are therefore prohibited, because $[0, 1]$ is not the projection of $[\mathbf{x}]$ onto the second axis. This guarantees the uniqueness of notation of a given box. The set of all n-dimensional boxes will be denoted by $\mathbb{IR}^n$. Non-empty boxes are n-dimensional axis-aligned parallelepipeds. Figure 2.4 illustrates the case $n = 2$, with $[\mathbf{x}] = [x_1] \times [x_2]$.

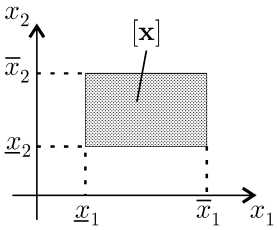


Fig. 2.4. A box $[\mathbf{x}]$ of $\mathbb{IR}^2$

Many of the notions introduced in Section 2.3.2 for intervals extend without difficulty to boxes. For instance, a box will be said to be punctual if *all* its interval components are. Any box with at least one punctual component

has a zero volume, so a box with a zero volume may not be punctual. The *lower bound* $\mathrm{lb}([\mathbf{x}])$ of a box $[\mathbf{x}]$ is the punctual vector consisting of the lower bounds of its interval components:

$$\underline{\mathbf{x}} = \ \mathrm{lb}([\mathbf{x}]) \triangleq (\mathrm{lb}([x_1]), \mathrm{lb}([x_2]), \cdots, \mathrm{lb}([x_n]))^{\mathrm{T}} = (\underline{x}_1, \underline{x}_2, \cdots, \underline{x}_n)^{\mathrm{T}}.$$

Similarly, the *upper bound* $\mathrm{ub}([\mathbf{x}])$ of $[\mathbf{x}]$ is the punctual vector

$$\overline{\mathbf{x}} = \mathrm{ub}([\mathbf{x}]) \triangleq (\mathrm{ub}([x_1]), \mathrm{ub}([x_2]), \cdots, \mathrm{ub}([x_n]))^{\mathrm{T}} = (\overline{x}_1, \overline{x}_2, \cdots, \overline{x}_n)^{\mathrm{T}}.$$

The *width* of $[\mathbf{x}] = ([x_1], [x_2], \ldots, [x_n])^{\mathrm{T}}$ is

$$w([\mathbf{x}]) \triangleq \max_{1 \leqslant i \leqslant n} w([x_i]). \tag{2.61}$$

If $[\mathbf{x}]$ is bounded and non-empty, then its *midpoint* (or *centre*) is

$$\mathrm{mid}([\mathbf{x}]) \triangleq (\mathrm{mid}([x_1]), \ldots, \mathrm{mid}([x_n]))^{\mathrm{T}}. \tag{2.62}$$

The *interval hull* of a subset $\mathbb{A}$ of $\mathbb{R}^n$ is the smallest box of $\mathbb{IR}^n$ that contains $\mathbb{A}$, denoted by $[\mathbb{A}]$.

The intersection of the boxes $[\mathbf{x}]$ and $[\mathbf{y}]$ of $\mathbb{IR}^n$ satisfies

$$[\mathbf{x}] \cap [\mathbf{y}] \triangleq ([x_1] \cap [y_1]) \times \ldots \times ([x_n] \cap [y_n]), \tag{2.63}$$

provided that $[\mathbf{x}] \cap [\mathbf{y}]$ is non-empty. Applying this rule for the intersection of $[1, 3] \times [-1, 2]$ and $[2, 4] \times [3, 7]$ would lead to $[2, 3] \times \varnothing$, which is actually correct but inconsistent with the convention that $[2, 3]$ should then be the projection of the result onto the first axis.

Most often the union of two boxes $[\mathbf{x}]$ and $[\mathbf{y}]$ is not a box. A favourable case is when there exists a unique $i \in \{1, \ldots, n\}$ such that $[x_j] = [y_j]$ for all $j \neq i$ and $[x_i] \cap [y_i] \neq \varnothing$, because then $[\mathbf{x}] \cup [\mathbf{y}] = ([x_1], \ldots, [x_{i-1}], [x_i] \cup [y_i], [x_{i+1}], \ldots, [x_n])^{\mathrm{T}}$. This seemingly unlikely event turns out to be frequently encountered with some of the algorithms to be considered in this book. In all other cases, the interval hull $[[\mathbf{x}] \cup [\mathbf{y}]]$ of the union of $[\mathbf{x}]$ and $[\mathbf{y}]$ can be computed as

$$[\mathbf{x}] \sqcup [\mathbf{y}] \triangleq ([x_1] \sqcup [y_1]) \times \ldots \times ([x_n] \sqcup [y_n]), \tag{2.64}$$

a process that extends to any number of boxes, some of which may be the empty set. We also have

$$[\mathbf{x}] \subset [\mathbf{y}] \Leftrightarrow [x_1] \subset [y_1] \text{ and } \ldots \text{ and } [x_n] \subset [y_n], \tag{2.65}$$

and

$$\mathbf{x} \in [\mathbf{y}] \Leftrightarrow x_1 \in [y_1] \text{ and } \ldots \text{ and } x_n \in [y_n]. \tag{2.66}$$

Classical operations for interval vectors are direct extensions of the same operations for punctual vectors. For instance, if $[\mathbf{x}]$ and $[\mathbf{y}]$ are boxes of $\mathbb{IR}^n$, and if α is a real number, then

$$\begin{aligned}
&\alpha[\mathbf{x}] \triangleq (\alpha[x_1]) \times \cdots \times (\alpha[x_n])\,,\\
&[\mathbf{x}]^{\mathrm{T}} * [\mathbf{y}] \triangleq [x_1] * [y_1] + \cdots + [x_n] * [y_n],\\
&[\mathbf{x}] + [\mathbf{y}] \triangleq ([x_1] + [y_1]) \times \cdots \times ([x_n] + [y_n])\,.
\end{aligned} \tag{2.67}$$

These definitions are consistent with the more general set operation of (2.11), page 13.

2.3.5 Interval matrices

Let $\mathbb{R}^{m\times n}$ be the set of all matrices with real coefficients, m rows and n columns. An $(m \times n)$-dimensional interval matrix is a subset of $\mathbb{R}^{m\times n}$ that can be defined as the Cartesian product of mn closed intervals. The interval matrix $[\mathbf{A}]$ will be written indifferently in any of the following forms:

$$\begin{aligned}
[\mathbf{A}] &= \begin{pmatrix} [a_{11}] & \dots & [a_{1n}] \\ \vdots & & \vdots \\ [a_{m1}] & \dots & [a_{mn}] \end{pmatrix}\\
&= [a_{11}] \times [a_{12}] \times \cdots \times [a_{mn}] = ([a_{ij}])_{1\leqslant i\leqslant m, 1\leqslant j\leqslant n}\,,
\end{aligned} \tag{2.68}$$

where $[a_{ij}] = [\underline{a}_{ij}, \overline{a}_{ij}]$ is the projection of $[\mathbf{A}]$ onto the (i,j)th axis. This convention makes unique the representation of the empty matrix. For instance,

$$[\mathbf{A}] = \begin{pmatrix} \varnothing & [0,1] \\ \varnothing & \varnothing \end{pmatrix} \tag{2.69}$$

is not allowed because the projection of the matrix represented by the Cartesian product $\varnothing \times [0,1] \times \varnothing \times \varnothing$ onto the $(1,2)$-axis is empty and not equal to $[0,1]$. The set of all $m \times n$ interval matrices is denoted by $\mathbb{IR}^{m\times n}$. An interval matrix is said to be punctual if all its entries are punctual. The *lower bound* $\mathrm{lb}([\mathbf{A}])$ of an interval matrix $[\mathbf{A}]$ is the punctual matrix made up with the lower bounds of its interval components:

$$\underline{\mathbf{A}} = \ \mathrm{lb}([\mathbf{A}]) \triangleq \begin{pmatrix} \underline{a}_{11} & \cdots & \underline{a}_{1n} \\ \vdots & & \vdots \\ \underline{a}_{m1} & \cdots & \underline{a}_{mn} \end{pmatrix}, \tag{2.70}$$

Similarly, its *upper bound* $\mathrm{ub}([\mathbf{A}])$ is the punctual matrix

$$\overline{\mathbf{A}} = \mathrm{ub}([\mathbf{A}]) \triangleq \begin{pmatrix} \overline{a}_{11} & \dots & \overline{a}_{1n} \\ \vdots & & \vdots \\ \overline{a}_{m1} & \dots & \overline{a}_{mn} \end{pmatrix}. \tag{2.71}$$

Its *width* $w([\mathbf{A}])$ is

$$w([\mathbf{A}]) \triangleq \max_{1\leqslant i\leqslant m, 1\leqslant j\leqslant n} w([a_{ij}]). \tag{2.72}$$

If $[\mathbf{A}] \in \mathbb{IR}^{m\times n}$ is bounded and non-empty, then its *midpoint* (or *centre*) is given by

$$\mathrm{mid}([\mathbf{A}]) = (\mathrm{mid}([a_{ij}]))_{1\leqslant i\leqslant m, 1\leqslant j\leqslant n}\,. \tag{2.73}$$

For $[\mathbf{A}]$ and $[\mathbf{B}]$ in $\mathbb{IR}^{m\times n}$ and $\mathbf{C}$ in $\mathbb{R}^{m\times n}$,

$$[\mathbf{A}] \subset [\mathbf{B}] \Leftrightarrow [a_{ij}] \subset [b_{ij}] \text{ for } 1 \leqslant i \leqslant m,\ 1 \leqslant j \leqslant n, \tag{2.74}$$

$$\mathbf{C} \in [\mathbf{B}] \Leftrightarrow c_{ij} \in [b_{ij}] \text{ for } 1 \leqslant i \leqslant m,\ 1 \leqslant j \leqslant n. \tag{2.75}$$

The *interval hull* of a set $\mathbb{A}$ of matrices of $\mathbb{R}^{n\times m}$ is the smallest element of $\mathbb{IR}^{n\times m}$ that contains $\mathbb{A}$.

If $[\mathbf{A}]$ and $[\mathbf{B}]$ are intervals, interval vectors or interval matrices of appropriate dimensions and if $\diamond$ is a binary operator, then

$$[\mathbf{A}] \diamond [\mathbf{B}] = [\{\mathbf{A} \diamond \mathbf{B} \mid \mathbf{A} \in [\mathbf{A}] \text{ and } \mathbf{B} \in [\mathbf{B}]\}]\,. \tag{2.76}$$

For instance, if $[\mathbf{A}]$ and $[\mathbf{B}]$ are in $\mathbb{IR}^{n\times n}$, $[\mathbf{x}]$ is in $\mathbb{IR}^n$ and α is in $\mathbb{R}$, then

$$\begin{array}{l} \alpha[\mathbf{A}] = (\alpha[a_{11}]) \times \cdots \times (\alpha[a_{nn}])\,, \\ [\mathbf{A}] + [\mathbf{B}] = ([a_{ij}] + [b_{ij}])_{1\leqslant i\leqslant n, 1\leqslant j\leqslant n}\,, \\ [\mathbf{A}] * [\mathbf{B}] = (\sum_{k=1}^{n} [a_{ik}] * [b_{kj}])_{1\leqslant i\leqslant n, 1\leqslant j\leqslant n}\,, \\ [\mathbf{A}] * [\mathbf{x}] = \left(\sum_{j=1}^{n} [a_{ij}] * [x_j]\right)_{1\leqslant i\leqslant n}. \end{array} \tag{2.77}$$

As with intervals, the product of two interval matrices will be denoted indifferently by $[\mathbf{A}] * [\mathbf{B}]$, or $[\mathbf{A}][\mathbf{B}]$. Some classical properties of matrices in a punctual context are no longer true. For instance, product is no longer associative

$$([\mathbf{A}][\mathbf{B}])[\mathbf{C}] \neq [\mathbf{A}]([\mathbf{B}][\mathbf{C}]), \tag{2.78}$$

or commutative with respect to scalars

$$[\alpha]([\mathbf{A}][\mathbf{x}]) \neq [\mathbf{A}]([\alpha][\mathbf{x}]). \tag{2.79}$$

Part of these specificities can be explained by the wrapping effect, as illustrated by the following example where it is shown that

$$\mathbf{A}[\mathbf{x}] \supset \{\mathbf{A}\mathbf{x} \mid \mathbf{x} \in [\mathbf{x}]\}. \tag{2.80}$$

Example 2.6 *Take*

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad [\mathbf{x}] = \begin{pmatrix} [-1, 0] \\ [1, 2] \end{pmatrix}. \tag{2.81}$$

Then

$$\mathbf{A}\,[\mathbf{x}] = \begin{pmatrix} [0, 2] \\ [1, 2] \end{pmatrix}, \tag{2.82}$$

which implies that $(0,2)^T$ *belongs to* $\mathbf{A}[\mathbf{x}]$*, whereas it does not belong to the actual value set* $\mathbb{B} = \{\mathbf{Ax} \mid \mathbf{x} \in [\mathbf{x}]\}$*, as Figure 2.5 makes clear.* ■

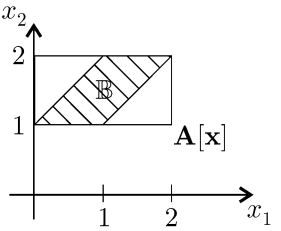


Fig. 2.5. Pessimism introduced by the wrapping effect

Of course, linear algebra also involves more sophisticated operations such as matrix inversion and the computation of eigenvalues and eigenvectors, which raise difficulties that go beyond this introductory chapter. See Neumaier (1990) for more details.

2.4 Inclusion Functions

2.4.1 Definitions

Consider a function $\mathbf{f}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$. The interval function $[\mathbf{f}]$ from $\mathbb{IR}^n$ to $\mathbb{IR}^m$ is an *inclusion function* for $\mathbf{f}$ if

$$\forall\, [\mathbf{x}] \in \mathbb{IR}^n, \quad \mathbf{f}\,([\mathbf{x}]) \subset [\mathbf{f}]\,([\mathbf{x}])\,. \tag{2.83}$$

The interval function $[\mathbf{f}]\,([\mathbf{x}]) = \mathbb{R}^m$, for all $[\mathbf{x}] \in \mathbb{IR}^n$, is an example of a (not very useful) inclusion function for all functions $\mathbf{f}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$. One of the purposes of interval analysis is to provide, for a large class of functions $\mathbf{f}$, inclusion functions that can be evaluated reasonably quickly and such that $[\mathbf{f}]\,([\mathbf{x}])$ is not too large. The function $\mathbf{f}$ may, for instance, be polynomial (Malan et al., 1992; Garloff, 2000), or given by an algorithm (Moore, 1979). It may even be defined as the solution of a set of differential equations (Lohner, 1987; Berz and Makino, 1998; Kühn, 1998).

To illustrate the notion of inclusion function, consider a function $\mathbf{f}$ from $\mathbb{R}^2$ to $\mathbb{R}^2$, with variables x_1 and x_2 that vary within $[x_1]$ and $[x_2]$. The image set $\mathbf{f}([\mathbf{x}])$ may have any shape. It may be non-convex (*i.e.*, there are points of $\mathbf{f}([\mathbf{x}])$ such that the line segment connecting them is not in $\mathbf{f}([\mathbf{x}])$), or even disconnected (*i.e.*, $\mathbf{f}([\mathbf{x}])$ is a union of disjoint sets) if $\mathbf{f}$ is discontinuous. Whatever the shape of $\mathbf{f}([\mathbf{x}])$, an inclusion function $[\mathbf{f}]$ of $\mathbf{f}$ makes it possible to compute a box $[\mathbf{f}]([\mathbf{x}])$ guaranteed to contain it (Figure 2.6).

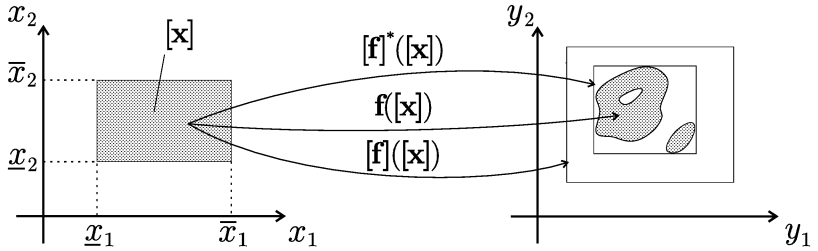


Fig. 2.6. Images of a box by a vector function $\mathbf{f}$ and two of its inclusion functions $[\mathbf{f}]$ and $[\mathbf{f}]^*$; $[\mathbf{f}]^*$ is minimal

Actually, as suggested by Figure 2.6, $[\mathbf{f}]([\mathbf{x}])$ may offer a very pessimistic vision of $\mathbf{f}([\mathbf{x}])$. But remarkable properties of $\mathbf{f}$, such as the positivity of some of its components, may be preserved by $[\mathbf{f}]$. Given that it is far easier to manipulate boxes than generic sets, this is a very interesting standpoint for the observation of a vector function.

An inclusion function $[\mathbf{f}]$ for $\mathbf{f}$ is *thin* if, for any punctual interval vector $[\mathbf{x}] = \mathbf{x}$, $[\mathbf{f}]([\mathbf{x}]) = \mathbf{f}(\mathbf{x})$. It is *convergent* if, for any sequence of boxes $[\mathbf{x}](k)$,

$$\lim_{k\to\infty} w([\mathbf{x}](k)) = 0 \Rightarrow \lim_{k\to\infty} w([\mathbf{f}]([\mathbf{x}](k))) = 0. \tag{2.84}$$

This property is illustrated by Figure 2.7. Note that if $[\mathbf{f}]$ is convergent, it is necessarily thin. The convergence of inclusion functions is required for proving the convergence of interval solvers such as those presented in Chapter 5. The inclusion function $[\mathbf{f}]$ is *minimal* if for any $[\mathbf{x}]$, $[\mathbf{f}]([\mathbf{x}])$ is the smallest box that contains $\mathbf{f}([\mathbf{x}])$. The minimal inclusion function for $\mathbf{f}$ is unique and will be denoted by $[\mathbf{f}]^*$ (see Figure 2.6).

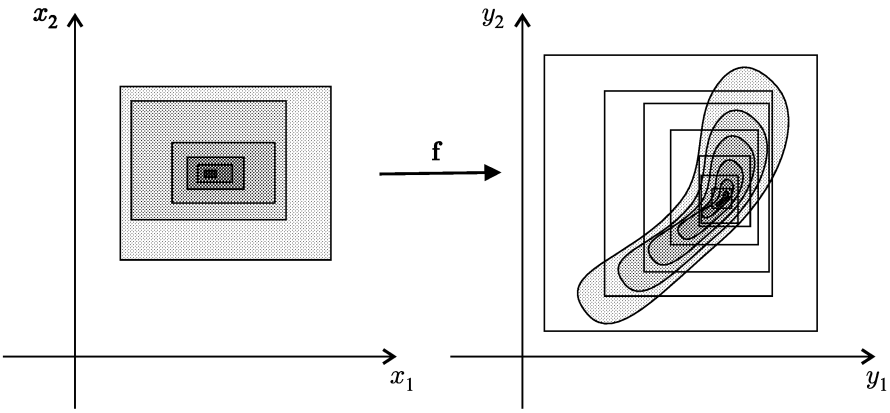


Fig. 2.7. A convergent inclusion function

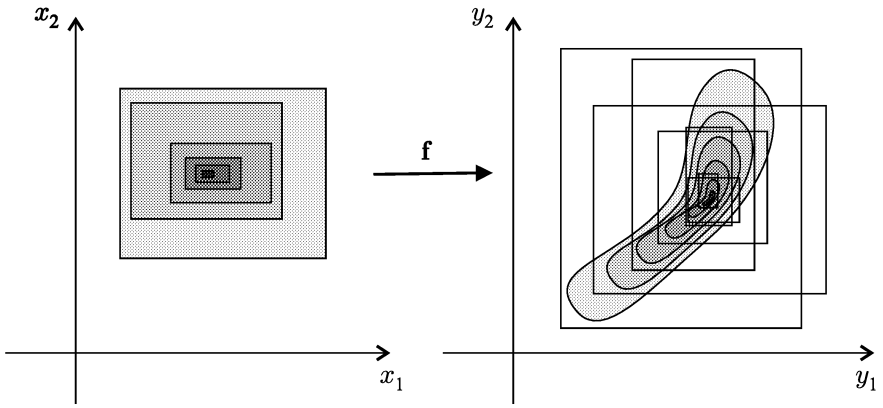


Fig. 2.8. A convergent inclusion function that is not inclusion monotonic

$[\mathbf{f}]$ is *inclusion monotonic* if

$$[\mathbf{x}] \subset [\mathbf{y}] \Rightarrow [\mathbf{f}]([\mathbf{x}]) \subset [\mathbf{f}]([\mathbf{y}]). \tag{2.85}$$

It is trivial to check that a minimal inclusion function is inclusion monotonic but not necessarily convergent (because $\mathbf{f}$ may be discontinuous). A convergent inclusion function may not be inclusion monotonic (see Figure 2.8).

Consider a function $\mathbf{f}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$, and let $[f_j]$, $j = 1, \ldots, m$, be m inclusion functions from $\mathbb{IR}^n$ to $\mathbb{IR}$ associated with the coordinate functions f_j of $\mathbf{f}$. An inclusion function for $\mathbf{f}$ is then given by

$$[\mathbf{f}]([\mathbf{x}]) = [f_1]([\mathbf{x}]) \times \cdots \times [f_m]([\mathbf{x}]). \tag{2.86}$$

$[\mathbf{f}]$ is convergent (thin, minimal, inclusion monotonic, respectively) if all its coordinate functions $[f_i]([\mathbf{x}])$ are convergent (thin, minimal, inclusion monotonic, respectively). The construction of inclusion functions for $\mathbf{f}$ can therefore be cast into that of inclusion functions for each of its coordinate functions. This is why we shall focus attention on getting inclusion functions for real-valued functions.

2.4.2 Natural inclusion functions

The first idea that comes to mind in order to build an inclusion function for $f : \mathbb{R}^n \to \mathbb{R}$ is to perform two optimizations to compute the *infimum* and *supremum* of f when each x_i is constrained to belong to $[x_i]$. At least in principle, one should thus get the smallest interval containing $f([x_1], [x_2], \ldots, [x_n])$, denoted by $[f]^*([x_1], [x_2], \ldots, [x_n])$. However, these optimization problems turn out to be far from trivial in general.

An alternative and much more tractable approach uses the following theorem, which is a direct consequence of Theorem 2.1, page 14.

Theorem 2.2 *Consider a function*

$$\begin{aligned} &f : \mathbb{R}^n \to \mathbb{R}, \\ &\quad (x_1, \ldots, x_n) \mapsto f(x_1, \ldots, x_n), \end{aligned} \tag{2.87}$$

expressed as a finite composition of the operators $+, -, *, /$ *and elementary functions (*$\sin, \cos, \exp,$ sqr$\ldots$*). An inclusion monotonic and thin inclusion function* $[f] : \mathbb{IR}^n \to \mathbb{IR}$ *for* f *is obtained by replacing each real variable* x_i *by an interval variable* $[x_i]$ *and each operator or function by its interval counterpart. This function is called the* natural inclusion function *of* f*. If* f *involves only continuous operators and continuous elementary functions, then* $[f]$ *is convergent. If, moreover, each of the variables* $(x_1, \ldots, x_n)$ *occurs at most once in the formal expression of* f *then* $[f]$ *is minimal.* ■

Remark 2.6 *Contrary to set computation, it is not sufficient that each input variable* x_i *appears at most once for the natural inclusion function to be minimal. Because of the wrapping effect, it is also required that all functions and operators involved in the expression of* f *be continuous. Consider, for instance, the continuous function* $f(x) = (\text{sign}(x))^2$*, where* $\text{sign}(x)$ *is equal to* 1 *if* $x \geqslant 0$ *and to* -1 *otherwise. Its natural inclusion function* $[f]$ *satisfies* $[f]([-1,1]) = [-1,1]^2 = [0,1]$*. Although* x *occurs only once in the formal expression of* $f(x)$*,* $[f]$ *is not minimal as* $f([-1,1]) = 1$*.* ■

Natural inclusion functions are not minimal in general, because of the dependency and wrapping effects. The accuracy of the resulting interval strongly depends on the expression of f, as illustrated by the three following examples. The first one presents a function of one variable to allow a graphical illustration. The next one involves a function of two variables. The last one shows how to deal with transcendental functions.

Example 2.7 *Consider the following four formal expressions of the same function* $f(x)$*:*

$$f_1(x) = x(x+1), \tag{2.88}$$

$$f_2(x) = x * x + x, \tag{2.89}$$

$$f_3(x) = x^2 + x, \tag{2.90}$$

$$f_4(x) = (x + \frac{1}{2})^2 - \frac{1}{4}. \tag{2.91}$$

Evaluate their natural inclusion functions for $[x] = [-1,1]$*:*

$$[f_1]([x]) = [x]([x]+1) = [-2,2], \tag{2.92}$$

$$[f_2]([x]) = [x] * [x] + [x] = [-2,2], \tag{2.93}$$

$$[f_3]([x]) = [x]^2 + [x] = [-1,2], \tag{2.94}$$

$$[f_4]([x]) = \left([x] + \frac{1}{2}\right)^2 - \frac{1}{4} = \left[-\frac{1}{4}, 2\right]. \tag{2.95}$$

The accuracy of the interval result thus depends on the formal expression of f (see Figure 2.9). Since $[x]$ occurs only once in f_4 and f_4 is continuous, $[f_4]$ is minimal. Thus $[f_4]([x]) = f([x]) = \left[-\frac{1}{4}, 2\right]$. ■

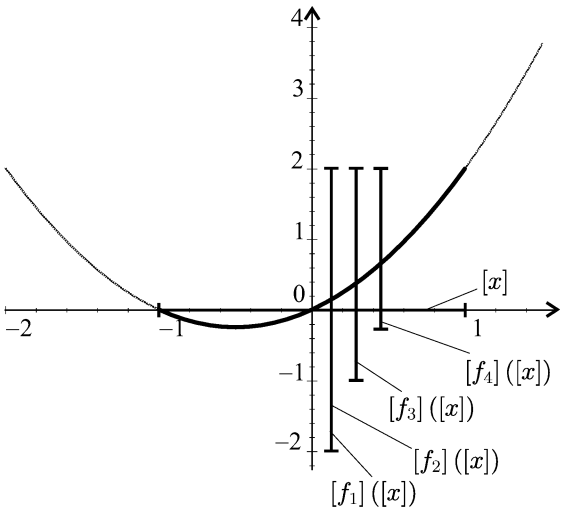


Fig. 2.9. Four natural inclusion functions for the same function

Example 2.8 *Consider the real function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by*

$$f(x_1, x_2) = \frac{x_1 - x_2}{x_1 + x_2}, \qquad \text{with } x_1 \in [-1, 2] \text{ and } x_2 \in [3, 5]. \tag{2.96}$$

The natural inclusion function $[f]_1$ for f is obtained by replacing each real variable by an interval variable, and each real operation by its interval counterpart:

$$[f]_1([x_1], [x_2]) = \frac{[x_1] - [x_2]}{[x_1] + [x_2]}, \tag{2.97}$$

so

$$\begin{aligned}[f]_1([-1, 2], [3, 5]) &= \frac{[-1, 2] - [3, 5]}{[-1, 2] + [3, 5]} = \frac{[-6, -1]}{[2, 7]} \\ &= [-6, -1] * [\frac{1}{7}, \frac{1}{2}] = [-3, -\frac{1}{7}].\end{aligned} \tag{2.98}$$

A second interval extension $[f]_2$ can similarly be obtained after rewriting f in such a way that x_1 and x_2 each appear only once:

$$[f]_2([x_1],[x_2]) = 1 - \frac{2}{1+[x_1]/[x_2]}. \tag{2.99}$$

Then

$$\begin{aligned}[f]_2([-1,2],[3,5]) &= 1 - \frac{2}{1+[-1,2]/[3,5]} = 1 - \frac{2}{1+[-1/3,2/3]} \\ &= 1 - \frac{2}{[2/3,5/3]} = 1 - [6/5,3] \\ &= [-2,-1/5]. \end{aligned} \tag{2.100}$$

$[f]_1$ and $[f]_2$ are both interval extensions of f. $[f]_2$ is more accurate than $[f]_1$, which suffers from the dependency effect. The interval computed by $[f]_2$ is minimal, and thus equal to the image set $f([-1,2],[3,5])$. ■

Example 2.9 *Consider the real function f defined by*

$$f(x_1,x_2) = \ln(e^{x_1} + \sin(x_2)), \tag{2.101}$$

with $x_1 \in [0,1]$ and $x_2 \in [\pi/4, 4\pi/3]$. Its natural inclusion function is

$$[f]([x_1],[x_2]) = \ln(\exp([x_1]) + \sin([x_2])), \tag{2.102}$$

so

$$\begin{aligned}[f]([0,1],[\pi/4,4\pi/3]) &= \ln(\exp([0,1]) + \sin([\pi/4,4\pi/3])) \\ &= \ln([1,e] + [-\sqrt{3}/2,1]) \\ &= \ln([1-\sqrt{3}/2, e+1]) \\ &= [\ln(1-\sqrt{3}/2), \ln(e+1)] \\ &\subset [-2.0101, 1.3133]. \end{aligned} \tag{2.103}$$

Since each variable appears only once, and since all the functions and operators involved in the formal expression of f are continuous, the penultimate interval is the exact image set $f([0,1],[\pi/4,4\pi/3])$. Note that the minimum and maximum values taken by f have thus been computed without performing a single optimization, although f is not monotonic. The last interval is a guaranteed numerical estimate of the previous one, obtained by outward rounding, see Chapter 10. ■

The use of natural inclusion functions is not always to be recommended, however. Their efficiency depends strongly on the number of occurrences of each variable, which is often difficult to reduce. An important field of investigation of interval analysis has thus been to propose other types of inclusion function that would provide less pessimistic results (Ratschek and Rokne, 1984), as shown in Sections 2.4.3 to 2.4.5.

2.4.3 Centred inclusion functions

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a scalar function of a vector $\mathbf{x} = (x_1, \ldots, x_n)^{\mathrm{T}}$. Assume that f is differentiable over the box $[\mathbf{x}]$, and denote $\mathrm{mid}([\mathbf{x}])$ by $\mathbf{m}$. The mean-value theorem then implies that

$$\forall \mathbf{x} \in [\mathbf{x}], \exists \mathbf{z} \in [\mathbf{x}] \mid f(\mathbf{x}) = f(\mathbf{m}) + \mathbf{g}^{\mathrm{T}}(\mathbf{z})(\mathbf{x} - \mathbf{m}), \tag{2.104}$$

where $\mathbf{g}$ is the *gradient* of f, *i.e.*, a column vector with entries $g_i = \partial f / \partial x_i$, $i = 1, \ldots, n$. Thus,

$$\forall \mathbf{x} \in [\mathbf{x}], f(\mathbf{x}) \in f(\mathbf{m}) + [\mathbf{g}^{\mathrm{T}}]([\mathbf{x}])(\mathbf{x} - \mathbf{m}), \tag{2.105}$$

where $[\mathbf{g}^{\mathrm{T}}]$ is an inclusion function for $\mathbf{g}^{\mathrm{T}}$, so

$$f([\mathbf{x}]) \subseteq f(\mathbf{m}) + [\mathbf{g}^{\mathrm{T}}]([\mathbf{x}])([\mathbf{x}] - \mathbf{m}). \tag{2.106}$$

Therefore, the interval function

$$[f_{\mathrm{c}}]([\mathbf{x}]) \triangleq f(\mathbf{m}) + [\mathbf{g}^{\mathrm{T}}]([\mathbf{x}])([\mathbf{x}] - \mathbf{m}) \tag{2.107}$$

is an inclusion function for $\mathbf{f}$, which we shall call the *centred inclusion function*. To illustrate the interest of this function in the one-dimensional case, consider the function $[f_{\mathrm{c}}](x)$ from $\mathbb{R}$ to $\mathbb{IR}$ defined by

$$[f_{\mathrm{c}}](x) \triangleq f(m) + [f']([x])(x - m) \tag{2.108}$$

for any given $[x]$. This function can be viewed as affine in x with an uncertain slope belonging to $[f']([x])$. The graph of $[f_{\mathrm{c}}](x)$ can thus be represented by a cone with centre $(m, f(m))$ as illustrated by Figure 2.10 for decreasing widths of $[x]$. The smaller $w([x])$ is, the better the cone approximates the function. The figure illustrates the fact that

$$\frac{w([f_{\mathrm{c}}]([x]))}{w(f([x]))} \to 1 \tag{2.109}$$

when the width of $[x]$ tends to 0, which is not the case in general for a natural inclusion function.

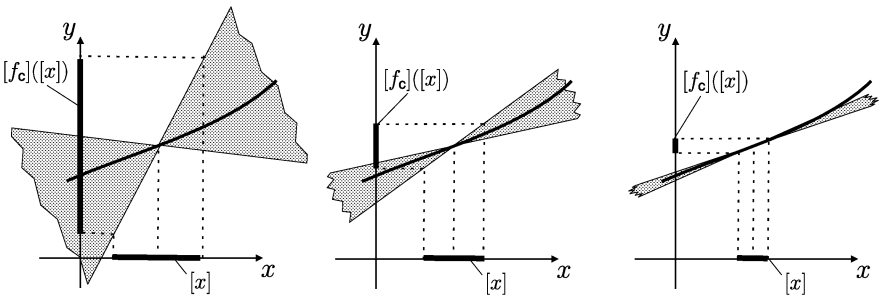


Fig. 2.10. Interpretation of the centred inclusion function

When the width of $[\mathbf{x}]$ is small, the effect of the pessimism possibly resulting from the interval evaluation of $[\mathbf{g}]([\mathbf{x}])$ is reduced by the scalar product with $[\mathbf{x}] - \mathbf{m}$, which is a small interval centred on zero.

2.4.4 Mixed centred inclusion functions

The centred inclusion function for a function f from $\mathbb{R}^n$ to $\mathbb{R}$ can be noticeably improved at the cost of a slightly more complicated formulation (Hansen, 1968). Recall that, for a function φ from $\mathbb{R}$ to $\mathbb{R}$,

$$\varphi(x) \in \varphi(m) + \varphi'([x])([x] - m), \tag{2.110}$$

where $m = \text{ mid}([x])$. The main idea to get a mixed centred inclusion function is to apply (2.110) n times, considering each variable of f in turn. The case $n = 3$ will be treated first, to simplify exposition. Consider $f(x_1, x_2, x_3)$ as a function of x_3 only and take $m_3 = \text{ mid}([x_3])$; (2.110) then implies that

$$f(x_1, x_2, x_3) \in f(x_1, x_2, m_3) + g_3(x_1, x_2, [x_3]) * ([x_3] - m_3). \tag{2.111}$$

Consider now $f(x_1, x_2, m_3)$ as a function of x_2 only and take $m_2 = \text{ mid}([x_2])$; (2.110) then yields

$$f(x_1, x_2, m_3) \in f(x_1, m_2, m_3) + g_2(x_1, [x_2], m_3) * ([x_2] - m_2). \tag{2.112}$$

Finally, consider $f(x_1, m_2, m_3)$ as a function of x_1 and take $m_1 = \text{ mid}([x_1])$; (2.110) then leads to

$$f(x_1, m_2, m_3) \in f(m_1, m_2, m_3) + g_1([x_1], m_2, m_3) * ([x_1] - m_1). \tag{2.113}$$

Combine these three equations to get

$$\begin{aligned} f(x_1, x_2, x_3) \in f(m_1, m_2, m_3) &+ g_1([x_1], m_2, m_3) * ([x_1] - m_1) \\ &+ g_2(x_1, [x_2], m_3) * ([x_2] - m_2) \\ &+ g_3(x_1, x_2, [x_3]) * ([x_3] - m_3). \end{aligned} \tag{2.114}$$

Thus

$$\begin{aligned} f([x_1], [x_2], [x_3]) \subset f(m_1, m_2, m_3) &+ g_1([x_1], m_2, m_3) * ([x_1] - m_1) \\ &+ g_2([x_1], [x_2], m_3) * ([x_2] - m_2) \\ &+ g_3([x_1], [x_2], [x_3]) * ([x_3] - m_3). \end{aligned} \tag{2.115}$$

This expression can be generalized for a function f of n variables. With $\mathbf{x} = (x_1, \ldots, x_n)^{\mathrm{T}}$ and $\mathbf{m} = \text{mid}([\mathbf{x}])$, one gets

$$f([\mathbf{x}]) \subset f(\mathbf{m}) + \sum_{i=1}^{n} [g_i]([x_1], \ldots, [x_i], m_{i+1}, \ldots, m_n) * ([x_i] - m_i), \tag{2.116}$$

and the right-hand side of (2.116) defines the *mixed centred inclusion function*. The main difference with (2.107) lies in the arguments of the gradient. In

(2.116), interval and punctual arguments are mixed, which allows pessimism to be decreased, as

$$[\mathbf{g}](\mathrm{mid}([\mathbf{x}]), [\mathbf{x}]) \subset [\mathbf{g}]([\mathbf{x}]). \tag{2.117}$$

2.4.5 Taylor inclusion functions

Iterating the reasoning that led to the centred inclusion function, one may think of using Taylor series expansion to approximate a function f from $\mathbb{R}^n$ to $\mathbb{R}$ at a higher order. This leads to the *Taylor inclusion function*. Consider a second-order expansion as an illustration:

$$[f]_{\mathrm{T}}([\mathbf{x}]) = f(\mathbf{m}) + \mathbf{g}^{\mathrm{T}}(\mathbf{m})([\mathbf{x}] - \mathbf{m}) + \frac{1}{2}([\mathbf{x}] - \mathbf{m})^{\mathrm{T}}[\mathbf{H}]([\mathbf{x}])([\mathbf{x}] - \mathbf{m}), \tag{2.118}$$

where $\mathbf{m} = \mathrm{mid}([\mathbf{x}])$, $\mathbf{g}$ is again the gradient of f and $[\mathbf{H}]([\mathbf{x}])$ is the interval *Hessian matrix*. The entry $[\mathbf{H}]_{ij}$ of $[\mathbf{H}]$ is an inclusion function of

$$h_{ij} = \begin{cases} \partial^2 f/\partial x_i^2 & \text{if } j = i \quad (i = 1, \ldots, n), \\ 2\partial^2 f/\partial x_i x_j & \text{if } j < i \quad (i = 1, \ldots, n), \\ 0 & \text{otherwise.} \end{cases} \tag{2.119}$$

A symmetric form of the Hessian matrix ($h_{ij} = \partial^2 f/\partial x_i x_j$ for all i and j) could also be used, but the resulting increase in the number of interval components in $[\mathbf{H}]([\mathbf{x}])$ would then lead to an increase in the pessimism of $[f]_{\mathrm{T}}$. Pessimism can be reduced by replacing $[\mathbf{H}]([\mathbf{x}])$ by a mixed expression $[\mathbf{H}](\mathrm{mid}([\mathbf{x}]), [\mathbf{x}])$, as was done for the gradient in the mixed centred form of Section 2.4.4.

When f has only one variable, the nth-order Taylor inclusion function is given by

$$\begin{aligned}[f]_{\mathrm{T}}([x]) = f(m) + f'(m)([x] - m) + \cdots + f^{n-1}(m)\frac{([x] - m)^{n-1}}{(n-1)!} \\ + [f^n]([x])\frac{([x] - m)^n}{n!}.\end{aligned} \tag{2.120}$$

Evaluation of the Taylor inclusion function of order n thus requires computation of the derivatives of f up to nth order, which may entail cumbersome manipulations.

2.4.6 Comparison

Under mild technical conditions (Moore, 1979), the natural, centred and Taylor inclusion functions are convergent. Roughly speaking, the *convergence rate* of a convergent inclusion function is the largest α such that

$$\exists \beta \mid w([f]([\mathbf{x}])) - w(f([\mathbf{x}])) \leqslant \beta w([\mathbf{x}])^{\alpha} \tag{2.121}$$

when $w([\mathbf{x}])$ tends to 0. When an inclusion function is minimal, its convergence rate is infinite. The convergence rate of a natural inclusion function is at least linear ($\alpha \geqslant 1$), whereas the convergence rate of a centred form is at least quadratic ($\alpha \geqslant 2$). The convergence rate of a Taylor inclusion function is also at least quadratic for any order $n \geqslant 2$. Quadratic convergence looks of course more interesting than linear convergence, but it should be remembered that it only means that more accurate results will be obtained in the case of infinitesimal boxes. Nothing similar can be said on the behaviour of these inclusion functions for boxes of a more realistic size. When the box involved is large, the natural inclusion function is generally more satisfactory than the centred inclusion function, whereas the latter performs better when the box is small, with the mixed version superior to the standard version.

No approach to building an inclusion function can claim to be uniformly the best, and a compromise between complexity and efficiency must often be struck. One may also use several inclusion functions and take the intersection of their image sets to get a better approximation of the image set of the original function.

Example 2.10 *Consider the function f defined by*

$$f(x) = x^2 + \sin(x), \tag{2.122}$$

and the intervals

$$[x] = [\frac{2\pi}{3}, \frac{4\pi}{3}] \text{ and } [y] = [\frac{99\pi}{100}, \frac{101\pi}{100}]. \tag{2.123}$$

We shall compare the approximations of $f([x])$ and $f([y])$ obtained when using the natural, centred, Taylor of order two and minimal inclusion functions, which will be respectively denoted by $[f]_{\rm n}$, $[f]_{\rm c}$, $[f]_{\rm T}$, and $[f]^*$. See also Exercise 11.10, page 318. The first three of these functions are given by

$$[f]_{\rm n}([x]) = [x]^2 + \sin([x]), \tag{2.124}$$

$$[f]_{\rm c}([x]) = f(\pi) + ([x] - \pi)[f']([x]), \tag{2.125}$$

$$[f]_{\rm T}([x]) = f(\pi) + ([x] - \pi)f'(\pi) + \frac{([x] - \pi)^2}{2}[f'']([x]), \tag{2.126}$$

with

$$f'(x) = 2x + \cos(x) \quad \text{and} \quad f''(x) = 2 - \sin(x). \tag{2.127}$$

The minimal inclusion function is trivial to evaluate after noticing that f is increasing over $[x]$ (and thus over $[y] \subset [x]$). So

$$[f]^*([x]) = [\ \underline{x}^2 + \sin(\underline{x}),\ \overline{x}^2 + \sin(\overline{x})]. \tag{2.128}$$

The results obtained by evaluating each of these inclusion functions over $[x]$ and $[y]$ are indicated in Table 2.2, where $\Delta([f]([x]))$ stands for the value of $w([f]([x])) - w(f([x]))$.

Table 2.2. Comparing inclusion functions

	$[x] = \left[\frac{2\pi}{3}, \frac{4\pi}{3}\right]$		$[y] = \left[\frac{99\pi}{100}, \frac{101\pi}{100}\right]$	
$[f]$	$[f]([x])$	$\Delta([f]([x]))$	$[f]([y])$	$\Delta([f]([y]))$
$[f]_{\mathrm{n}}$	$[3.52046, 18.41199]$	3.46410	$[9.64178, 10.09940]$	0.12564
$[f]_{\mathrm{c}}$	$[1.62022, 18.11899]$	5.07134	$[9.70163, 10.03758]$	0.00397
$[f]_{\mathrm{T}}$	$[4.33706, 16.97362]$	1.20913	$[9.70362, 10.03659]$	0.00099
$[f]^*$	$[5.25251, 16.67994]$	0	$[9.70461, 10.03658]$	0

The numerical values are given with an accuracy of 10^{-5}*. It turns out that the natural inclusion function remains competitive for the larger interval* $[x]$*, which is an additional incentive to use it, besides its simplicity. The centred and Taylor inclusion functions are more efficient than the natural inclusion function for the smaller interval* $[y]$*. The Taylor inclusion function brings a noticeable improvement compared to the natural and centred inclusion functions, even for the larger interval. Finally, this example reminds us that it may be useful to check whether the function considered is monotonic, in which case obtaining a minimal form is trivial. Unfortunately, the expressions encountered in engineering applications are seldom as cooperative as the one considered here, which limits the practical interest of this remark.* ■

Example 2.11 *Consider now a vector function* $\mathbf{f}$ *from* $\mathbb{R}^2$ *to* $\mathbb{R}^2$*, defined by*

$$\begin{aligned} f_1(x_1, x_2) &= x_1^2 + x_1 \exp(x_2) - x_2^2, \\ f_2(x_1, x_2) &= x_1^2 - x_1 \exp(x_2) + x_2^2, \end{aligned} \tag{2.129}$$

where x_1 *and* x_2 *belong to* $[x_1]$ *and* $[x_2]$ *respectively. The natural inclusion function* $[\mathbf{f}]_{\mathrm{n}}$ *for* $\mathbf{f}$ *is given by*

$$[\mathbf{f}]_{\mathrm{n},1}([\mathbf{x}]) = [x_1]^2 + [x_1] \exp([x_2]) - [x_2]^2, \tag{2.130}$$

$$[\mathbf{f}]_{\mathrm{n},2}([\mathbf{x}]) = [x_1]^2 - [x_1] \exp([x_2]) + [x_2]^2. \tag{2.131}$$

The centred inclusion function $[\mathbf{f}]_{\mathrm{c}}$ *is given by*

$$[\mathbf{f}]_{\mathrm{c}}([\mathbf{x}]) = \mathbf{f}(\mathrm{mid}([\mathbf{x}])) + [\mathbf{J_f}]([\mathbf{x}]) * ([\mathbf{x}] - \mathrm{mid}([\mathbf{x}])). \tag{2.132}$$

All the arguments of the interval Jacobian matrix $[\mathbf{J_f}]$ *are intervals and the* i*th row of* $[\mathbf{J_f}]([\mathbf{x}])$ *is given by* $[\mathbf{g}_i^{\mathrm{T}}]([\mathbf{x}])$*, with* $\mathbf{g}_i$ *the gradient of the* i*th component of* $\mathbf{f}$ *(see (2.107) page 33):*

$$[\mathbf{J_f}]([\mathbf{x}]) = \begin{pmatrix} 2[x_1] + \exp([x_2]) & -2[x_2] + [x_1] \exp([x_2]) \\ 2[x_1] - \exp([x_2]) & 2[x_2] - [x_1] \exp([x_2]) \end{pmatrix}. \tag{2.133}$$

The mixed centred inclusion function $[\mathbf{f}]_{\mathrm{m}}$ *is given by*

$$[\mathbf{f}]_{\mathrm{m}}([\mathbf{x}]) = \mathbf{f}(\mathrm{mid}([\mathbf{x}])) + [\mathbf{J_f}](\mathrm{mid}([\mathbf{x}]), [\mathbf{x}]) * ([\mathbf{x}] - \mathrm{mid}([\mathbf{x}])), \tag{2.134}$$

where $[\mathbf{J_f}]$ *now depends on a mixture of punctual and interval arguments*

$$[\mathbf{J_f}](\text{mid}([\mathbf{x}]),[\mathbf{x}]) = \begin{pmatrix} 2[x_1]+\exp(\text{mid}([x_2])) & -2[x_2]+[x_1]\exp([x_2]) \\ 2[x_1]-\exp(\text{mid}([x_2])) & 2[x_2]-[x_1]\exp([x_2]) \end{pmatrix}. \tag{2.135}$$

As expected, (2.135) contains less interval arguments than (2.133). Table 2.3 compares the efficiency of the natural, centred and mixed centred inclusion functions on two boxes $[\mathbf{x}]$ *and* $[\mathbf{y}]$*. The definition of* Δ *is as in Example 2.10. The conclusions are similar to those of the scalar case: for the larger box* $[\mathbf{x}]$*, the natural inclusion function is more satisfactory, whereas the centred inclusion function performs better for the smaller box* $[\mathbf{y}]$*, with the mixed version superior to the non-mixed one.* ■

Table 2.3. Comparison of vector inclusion functions

	$[x_1]=[0.5,1.5]\,;[x_2]=[1.5,2.5]$		$[y_1]=[0.9,1.1]\,;[y_2]=[1.9,2.1]$	
$[\mathbf{f}]$	$[\mathbf{f}]([\mathbf{x}])$	$\Delta[[\mathbf{f}]([\mathbf{x}])]$	$[\mathbf{f}]([\mathbf{y}])$	$\Delta[[\mathbf{f}]([\mathbf{y}])]$
$[f_1]_{\text{n}}$	$[-3.75916, 18.2737]$	7.67904	$[2.41730, 6.58279]$	1.60000
$[f_2]_{\text{n}}$	$[-15.7737, 6.25916]$	11.67904	$[-4.56279, -0.39730]$	2.40001
$[f_1]_{\text{c}}$	$[-10.8391, 19.6172]$	16.10248	$[2.83416, 5.94395]$	0.54430
$[f_2]_{\text{c}}$	$[-15.6172, 10.8391]$	16.10248	$[-3.54395, -1.23416]$	0.54431
$[f_1]_{\text{m}}$	$[-8.44234, 17.2205]$	11.30902	$[2.91187, 5.86624]$	0.38888
$[f_2]_{\text{m}}$	$[-13.2205, 8.44234]$	11.30902	$[-3.46624, -1.31187]$	0.38889
$[f_1]^*$	$[-0.08008, 14.27374]$	0	$[3.21730, 5.78279]$	0
$[f_2]^*$	$[-9.77374, 0.58008]$	0	$[-3.36279, -1.59731]$	0

2.5 Inclusion Tests

Inclusions tests can be used to prove that all the points in a given box satisfy a given property, or to prove that none of them does. These tests involve interval Booleans, which will be presented first.

2.5.1 Interval Booleans

Set computation as defined in Section 2.2 can be used for the Boolean set

$$\mathbb{B} \triangleq \{\mathit{false}, \mathit{true}\}, \tag{2.136}$$

but there is no need to use wrappers for outer approximating the Boolean sets to be handled, since $\mathbb{B}$ is finite. There will thus be no wrapping effect, but

the dependency effect will still be present. A *Boolean number* is an element of $\mathbb{B}$. By extension[1], an *interval Boolean* is a subset of $\mathbb{B}$. Thus, the set of all interval Booleans is

$$\mathbb{IB} = \{\varnothing, 0, 1, [0,1]\}, \tag{2.137}$$

where $\varnothing$ stands for *impossible,* 0 for *false*, 1 for *true*, and $[0,1]$ for *indeterminate.* Operations on interval Booleans are easily defined in the framework of set computation:

$$\begin{aligned}
[a] \vee [b] &= \{a \vee b \mid a \in [a], b \in [b]\}, \\
[a] \wedge [b] &= \{a \wedge b \mid a \in [a], b \in [b]\}, \\
\neg [a] &= \{\neg a \mid a \in [a]\}, \\
[a] \cap [b] &= \{\max(\underline{a}, \underline{b}), \min(\overline{a}, \overline{b})\}, \\
[a] \cup [b] &= \{\min(\underline{a}, \underline{b}), \max(\overline{a}, \overline{b})\},
\end{aligned} \tag{2.138}$$

where $\wedge$ and $\vee$ respectively stand for the `AND` and `OR` operators and where $\neg$ is the complementation operator, such that$\neg 0 = 1$ and $\neg 1 = 0$. For instance,

$$([0,1] \vee 1) \wedge ([0,1] \wedge 1) = 1 \wedge [0,1] = [0,1]. \tag{2.139}$$

If $[a] \in \mathbb{IB}$, then

$$0 \wedge [a] = 0;\ 1 \wedge [a] = [a];\ [a] \wedge [a] = [a]; \tag{2.140}$$

$$0 \vee [a] = [a];\ 1 \vee [a] = 1;\ [a] \vee [a] = [a]. \tag{2.141}$$

The dependency effect is still present when the complementation operator is used. For instance,

$$[a] \subset ([a] \wedge [b]) \vee ([a] \wedge \neg [b]), \tag{2.142}$$

and the values of the two sides of (2.142) differ for $a = 0$ (or 1) and $[b] = [0,1]$, whereas $a = (a \wedge b) \vee (a \wedge \neg b)$.

Any function β from $\mathbb{B}^n$ to $\mathbb{B}$ will be called a *Boolean function*. The notion of inclusion function developed for real functions readily extends to Boolean functions. $[\beta] : \mathbb{IB}^n \to \mathbb{IB}$ is an *inclusion function* for β if

$$\forall ([b_1], \ldots, [b_n]) \in \mathbb{IB}^n, \beta([b_1], \ldots, [b_n]) \subset [\beta]([b_1], \ldots, [b_n]). \tag{2.143}$$

The *natural inclusion function* $[\beta]$ of β is obtained by replacing all arguments and operators of β by their interval counterparts. $[\beta]([b_1], \ldots, [b_n])$ is *minimal* if

$$\forall ([b_1], \ldots, [b_n]) \in \mathbb{IB}^n, \beta([b_1], \ldots, [b_n]) = [\beta]([b_1], \ldots, [b_n]). \tag{2.144}$$

[1] For any set equipped with a partial ordering $(\mathbb{S}, \leqslant)$, one can always define the set $\mathbb{IS}$, of the pairs $[a,b]$ such that $a \in \mathbb{S}$, $b \in \mathbb{S}$ and $a \leqslant b$. The elements of $\mathbb{IS}$ will be called intervals. $\mathbb{S}$ may for instance stand for $\mathbb{R}^n$, for the set of all Boolean numbers $\mathbb{B}$ or for the set of all compact sets. In the last case, the partial ordering is $\subset$.

As with real functions, whenever the expression $\beta(b_1, \ldots, b_n)$ is non-decreasing with respect to all its variables, the minimal inclusion function is given by

$$[\beta]^*([b_1], \ldots, [b_n]) = \left[\beta(\underline{b}_1, \ldots, \underline{b}_n), \beta(\overline{b}_1, \ldots, \overline{b}_n)\right]. \tag{2.145}$$

Note that this is a frequently encountered situation. It is the case, for instance, when the complementation operator is not used, *i.e.*, when β is a polynomial. There are nevertheless many Boolean expressions that are not monotonic, such as the *exclusive or*

$$\beta(b_1, b_2) = (b_1 \wedge \neg b_2) \vee (\neg b_1 \wedge b_2). \tag{2.146}$$

Even when a Boolean expression is not monotonic, it is always possible, at least in principle, to evaluate a minimal inclusion function for it, because each of the interval Booleans may take four values at most. Consider, for example, $\beta([b_1], [b_2])$, with $\beta(b_1, b_2) = (b_1 \wedge b_2) \vee (b_1 \wedge \neg b_2)$, for $[b_1] = 1$ and $[b_2] = [0, 1]$. The natural interval extension of β yields $[\beta](1, [0, 1]) = (1 \wedge [0, 1]) \vee (1 \wedge \neg [0, 1]) = [0, 1]$. A minimal evaluation is obtained by writing $\beta(1, [0, 1]) = \beta(1, 0) \cup \beta(1, 1) = 1$. Of course, this approach leads to a combinatorial explosion when the number of variables increases, which can sometimes be avoided by manipulating the Boolean expressions, for instance with the help of Karnaugh tables or by taking advantage of well-known simplification rules to reduce the number of occurrences of the Boolean variables.

2.5.2 Tests

A *test* is a function t from $\mathbb{R}^n$ to $\mathbb{B}$. An *inclusion test* for t is a function $[t]$ from $\mathbb{IR}^n$ to $\mathbb{IB}$ such that for any $[\mathbf{x}] \in \mathbb{IR}^n$,

$$\begin{array}{l} ([t]([\mathbf{x}]) = 1) \Rightarrow (\forall \mathbf{x} \in [\mathbf{x}],\ t(\mathbf{x}) = 1), \\ ([t]([\mathbf{x}]) = 0) \Rightarrow (\forall \mathbf{x} \in [\mathbf{x}],\ t(\mathbf{x}) = 0). \end{array} \tag{2.147}$$

An inclusion test $[t]$ is *thin* if $[t](\mathbf{x}) = t(\mathbf{x})$ for any $\mathbf{x} \in \mathbb{R}^n$. It is *minimal* if

$$\forall [\mathbf{x}] \in \mathbb{IR}^n, [t]([\mathbf{x}]) = \{t(\mathbf{x}) \mid \mathbf{x} \in [\mathbf{x}]\}. \tag{2.148}$$

A minimal test is necessarily thin.

Example 2.12 *Consider the test*

$$\begin{array}{rccc} t: & \mathbb{R}^2 & \to & \{0, 1\} \\ & (x_1, x_2)^T & \mapsto & (x_1 + x_2 \leqslant 5), \end{array} \tag{2.149}$$

which means that

$$t(\mathbf{x}) = \begin{cases} 1 \text{ if } x_1 + x_2 \leqslant 5, \\ 0 \text{ if } x_1 + x_2 > 5. \end{cases} \tag{2.150}$$

The minimal inclusion test $[t]$ *associated with* t *is given by*

$$[t]([\mathbf{x}]) = \begin{cases} 1 & \text{if } \overline{x}_1 + \overline{x}_2 \leqslant 5, \\ 0 & \text{if } \underline{x}_1 + \underline{x}_2 > 5, \\ [0,1] & \text{otherwise,} \end{cases} \tag{2.151}$$

which can be written more concisely as

$$[t]([\mathbf{x}]) \Leftrightarrow ([x_1] + [x_2] \leqslant 5)\,. \tag{2.152}$$

It is minimal and thin. ■

Any Boolean operator on real numbers, such as $(\leqslant, \geqslant, <, >$, integer, odd, even, prime...$)$ can be similarly extended to intervals. For instance,

$$\begin{aligned} &([a,b] \leqslant [c,d]) = 1 \quad \text{if } b \leqslant c, \\ &([a,b] \leqslant [c,d]) = 0 \quad \text{if } a > d, \\ &([a,b] \leqslant [c,d]) = [0,1] \text{ if neither } b \leqslant c \text{ nor } a > d. \end{aligned} \tag{2.153}$$

The Boolean comparison operator $=$ cannot be extended in this way, because it has already been given a bivalued meaning by set theory:

$$\begin{aligned} ([a,b] = [c,d]) &= 1 \text{ if } a = c \text{ and } b = d, \\ &= 0 \text{ otherwise.} \end{aligned} \tag{2.154}$$

With the help of interval analysis and the notion of inclusion function, it is easy to build an inclusion test for any test that can be put in the form

$$t(\mathbf{x}) = \beta(t_1(\mathbf{x}), \ldots, t_n(\mathbf{x})), \tag{2.155}$$

with

$$t_i(\mathbf{x}) \Leftrightarrow (f_i(\mathbf{x}) \geqslant 0)\,, \quad i = 1, \ldots, n, \tag{2.156}$$

and $\beta : \mathbb{B}^n \to \mathbb{B}$ a Boolean expression. This inclusion test is given by

$$[t]([\mathbf{x}]) = [\beta]([t_1]([\mathbf{x}]), \ldots, [t_n]([\mathbf{x}])), \tag{2.157}$$

with

$$[t_i]([\mathbf{x}]) \Leftrightarrow ([f_i]([\mathbf{x}]) \geqslant 0)\,, \quad i = 1, \ldots, n, \tag{2.158}$$

and $[\beta]$ some inclusion function for β. Note that even if the tests $[t_i]([\mathbf{x}])$ are all minimal and if $[\beta]$ is minimal too, the dependency effect is still lurking, so pessimism can still be introduced. For instance, the test $t(x) = (x \leqslant 7) \vee (x \geqslant 6)$ admits the inclusion test $[t]([x]) = ([x] \leqslant 7) \vee ([x] \geqslant 6)$. Despite the fact that $[t]([x])$ consists of two minimal inclusion tests and a polynomial and thus increasing expression $\beta(b_1, b_2) = b_1 \vee b_2$, this inclusion test is pessimistic. For instance for $[x] = [5,8]$,

$$[t]([x]) = ([5,8] \leqslant 7) \vee ([5,8] \geqslant 6) = [0,1] \ \vee \ [0,1] = [0,1], \tag{2.159}$$

whereas $t([x]) = \{t(x) \mid x \in [5,8]\} = 1$.

2.5.3 Inclusion tests for sets

Let $\mathbb{A}$ be a set of $\mathbb{R}^n$; an *inclusion test* $[t_{\mathbb{A}}]$ *for* $\mathbb{A}$ is an inclusion test for the test $t_{\mathbb{A}}(\mathbf{x}) \Leftrightarrow (\mathbf{x} \in \mathbb{A})$, *i.e.*, $[t_{\mathbb{A}}]$ satisfies

$$\begin{array}{l} [t_{\mathbb{A}}]([\mathbf{x}]) = 1 \Rightarrow (\forall \mathbf{x} \in [\mathbf{x}]\,,\ t_{\mathbb{A}}(\mathbf{x}) = 1) \Leftrightarrow ([\mathbf{x}] \subset \mathbb{A})\,, \\ [t_{\mathbb{A}}]([\mathbf{x}]) = 0 \Rightarrow (\forall \mathbf{x} \in [\mathbf{x}]\,,\ t_{\mathbb{A}}(\mathbf{x}) = 0) \Leftrightarrow ([\mathbf{x}] \cap \mathbb{A} = \varnothing)\,. \end{array} \tag{2.160}$$

When $[t_{\mathbb{A}}]([\mathbf{x}]) = [0,1]$, nothing can be concluded as to the inclusion of $[\mathbf{x}]$ in $\mathbb{A}$.

The notion of inclusion test for sets will simplify the presentation of algorithms in future chapters.

Previous definitions can be adapted to inclusion tests for sets:

$[t_{\mathbb{A}}]([\mathbf{x}])$ is *inclusion monotonic* iff	$([\mathbf{x}] \subset [\mathbf{y}]) \Rightarrow ([t_{\mathbb{A}}]([\mathbf{x}]) \subset [t_{\mathbb{A}}]([\mathbf{y}]))$
$[t_{\mathbb{A}}]$ is *minimal* iff	$\forall [\mathbf{x}] \in \mathbb{IR}^n,\ [t_{\mathbb{A}}]([\mathbf{x}]) = t_{\mathbb{A}}([\mathbf{x}])$
$[t_{\mathbb{A}}]$ is *thin* iff	$\forall \mathbf{x} \in \mathbb{R}^n,\ [t_{\mathbb{A}}](\mathbf{x}) \neq [0,1]$

The inclusion test $[t_{\mathbb{A}}]$ will be said to be *more accurate* than the inclusion test $[t'_{\mathbb{A}}]$ iff

$$\forall [\mathbf{x}] \in \mathbb{IR}^n, [t_{\mathbb{A}}]([\mathbf{x}]) \subset [t'_{\mathbb{A}}]([\mathbf{x}])\,. \tag{2.161}$$

The following properties can be used to build inclusion tests for sets defined from elementary set operations such as union, intersection or complementation. If $[t_{\mathbb{A}}]([\mathbf{x}])$ and $[t_{\mathbb{B}}]([\mathbf{x}])$ are thin inclusion tests for the sets $\mathbb{A}$ and $\mathbb{B}$, define

$$\begin{array}{ll} [t_{\mathbb{A}\cap\mathbb{B}}]([\mathbf{x}]) & \triangleq ([t_{\mathbb{A}}] \cap [t_{\mathbb{B}}])([\mathbf{x}]) = [t_{\mathbb{A}}]([\mathbf{x}]) \cap [t_{\mathbb{B}}]([\mathbf{x}]), \\ [t_{\mathbb{A}\cup\mathbb{B}}]([\mathbf{x}]) & \triangleq ([t_{\mathbb{A}}] \cup [t_{\mathbb{B}}])([\mathbf{x}]) = [t_{\mathbb{A}}]([\mathbf{x}]) \cup [t_{\mathbb{B}}]([\mathbf{x}]), \\ [t_{\neg\mathbb{A}}]([\mathbf{x}]) & \triangleq \neg\,[t_{\mathbb{A}}]([\mathbf{x}]) = 1 - [t_{\mathbb{A}}]([\mathbf{x}]). \end{array} \tag{2.162}$$

$[t_{\mathbb{A}\cap\mathbb{B}}]$, $[t_{\mathbb{A}\cup\mathbb{B}}]$ and $[t_{\neg\mathbb{A}}]$ are then thin inclusion tests for the sets $\mathbb{A} \cap \mathbb{B}$, $\mathbb{A} \cup \mathbb{B}$ and $\neg\mathbb{A} \triangleq \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \notin \mathbb{A}\}$, respectively.

2.6 Conclusions

Computation on sets should in general be viewed as an idealization, because generic sets cannot be represented and handled exactly by a computer. This is why the notion of set of wrappers was introduced. Wrappers are simple sets easily manipulated by computers, to be used to approximate more complicated sets.

Interval analysis uses intervals and boxes as wrappers, which makes it possible to compute outer approximations of ranges of functions. When evaluated for interval arguments, inclusion functions yield outer approximations

of the actual image sets of interest. Interval computation is thus usually pessimistic. This pessimism is due to the *dependency and wrapping effects*. The dependency effect, already present when computing on sets, takes place when variables occur several times in the formal expression of the function to be evaluated. The wrapping effect is due to the fact that generic sets are contained in intervals or boxes.

Pessimism may be reduced by transforming the formal expression of functions in order to decrease the number of occurrences of the variables or by using more sophisticated inclusion functions than the natural inclusion function obtained by replacing each operator and elementary function by its interval counterpart.

Intervals and boxes alone cannot describe all sets of interest with sufficient accuracy. The next chapter will show how this can be done by using unions of intervals or of boxes.